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European Commission

Annual Activity Report 2021

DG DEFENCE INDUSTRY AND SPACE

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THE DG IN BRIEF

DG DEFIS Mission Statement
DG Defence industry and space: Making EU more secure, sustainable and resilient

The Directorate-General for Defence industry and Space (DG DEFIS) develops and carries out the Commission's policies on defence industry and space.

EU competences and obligations for defence industry action are conferred under the TFEU Title XVII Industry, and in particular Article 173(3), and Title XIX Research and development, and in particular on Article 182(4), Article 183 and the second paragraph of Article 188 thereof; and for space, under Article 189 TFEU and Article 4 (3)TFEU.

DG DEFIS is responsible for a total budget of EUR 24.74 billion (for the period 2021-2027) through:

- the **European Defence Fund (EDF)** and its precursors the **Industrial Development Programme (EDIDP)** and the **Preparatory Action on Defence Research (PADR)** largely implemented through direct management by the Commission (mainly grants);
- the **EU Space Programme** implemented through direct management by the Commission or through indirect management by the European Union Agency for the Space Programme (EUSPA), the European Space Agency (ESA) and other entrusted entities;
- **Horizon Europe** for Space, managed directly by DG DEFIS and the Health and Digital Executive Agency (HaDEA) in close cooperation with DG RTD, as well as through indirect management with EUSPA and ESA.

In 2021, a major achievement was the adoption by the European Parliament and the Council of the Regulation of the new EU Space Programme and of the European Defence Fund and the start of their implementation.

DG DEFIS is organised around 3 directorates, 12 units and a ‘security of information task force’, with a total of 239 staff at the end of 2021.

Given that the majority of DG DEFIS budget is managed indirectly via entrusted entities, the DG relies on external control systems. These external control systems were validated by external independent auditors that performed pillar assessments (based on the common terms of reference of the Commission). These assessments and the implementation of their critical recommendations were a pre-requisite before the signature of contribution agreements with entrusted entities. This allows DG DEFIS to rely on the external control systems of these entities, which becomes part of the accountability chain of the DG.

In addition to the inherent risks related to the direct and indirect spending modes, DG DEFIS has to take account of other risks related to factors that are beyond the control of the DG, such as technical risks (space activity), extraordinary events as seen with the COVID-19 pandemic or Brexit, and security risks not under the full control of the DG.

EXECUTIVE SUMMARY

This Annual Activity Report is a management report of the Director-General of DG DEFIS to the College of Commissioners. Annual Activity Reports are the main instrument of management accountability within the Commission and constitute the basis on which the College takes political responsibility for the decisions it takes as well as for the coordinating, executive and management functions it exercises, as laid down in the Treaties ([1]).

A. Key results and progress towards achieving the Commission's general objectives and DG's specific objectives

The work of DG DEFIS in 2021 followed the [2021 Management Plan] and the intervention logic set out in the [Strategic Plan 2020-2024] to achieve the specific objectives of DG DEFIS and to contribute to the achievement of the general objectives of the Von der Leyen Commission. DG DEFIS programmes contributed in particular in making the EU more secure, resilient and sustainable whilst reinforcing the EU's strategic autonomy.

The major achievement of DG DEFIS in 2021 came with the adoption by the Council and European Parliament of the European Defence Fund Regulation and the EU Space Programme Regulation in April, after long and intensive negotiations in the past years. This allowed for the roll out of the new EU Space Programme (EUR 14.8 billion) and the European Defence Fund (EUR 7.9 billion) under the 2021-2027 EU budget period.

Overall, by starting the implementation of these two flagship programmes and ensuring the continuity of the activities in the previous programmes, DG DEFIS achieved its policy goals in 2021. With hindsight, the 2021 achievements constituted an excellent basis for the quick response in 2022 in the aftermath of the Russian aggression of Ukraine.

Tailored internal and external communication activities were deployed for each specific objective, making sure to target society and communities at large. Online presence and a range of 'virtual' events underpinned the success of many actions and contributed to strengthening the cross-sectoral uptake, increasing awareness about DG DEFIS portfolios and ensuring the presence of the EU as a credible actor at the global stage.

A European Green Deal

DG DEFIS is committed to making the EU more sustainable, including by maximising the take-up of EU space-enabled services in cross-policy EU legislative initiatives.

Copernicus, the EU Earth Observation and monitoring system, continued throughout 2021 to provide reliable Earth monitoring data and information, supporting the EU goals for 2050 climate-neutrality and sustainability. The number of users of the Copernicus services is steadily increasing, in particular the Copernicus climate change service (C3S) which has

([1]) Article 17(1) of the Treaty on European Union reached almost 100 000 users by end of 2021. The added value of the C3S services and the Copernicus Atmosphere Monitoring Service (CAMS) were given centre stage at the United Nations Climate Change Conference of the Parties (COP26) in November 2021, in Glasgow (UK).

Supporting the transition to carbon neutrality, Galileo and EGNOS, the EU satellite navigation systems, provided throughout 2021 high accuracy positioning and navigation signals that are essential for determining optimal routes in any mode of transport and lead to the reduction of CO_2 emissions. When it comes to air transport, positive results are shown by the increasing number of airports that are using EGNOS-based landing procedures which reached a total of 426 airports by end 2021.

DG DEFIS intensified the dialogue with the civil aeronautics industry and launched a public consultation that focused on the entry into service of hydrogen-powered and fully electric aircraft which emit no CO_2. The results of the consultation reflected the growing momentum in industry to bring such aircraft to market and confirmed the need for a cross-sectoral, flanking initiative such as an Alliance for Zero Emission Aviation to prepare the air transport system for their operation.

The EU Space Programme made an important contribution to the EU taxonomy Regulation and in defining the qualification of economic activities as environmentally sustainable.

A Europe fit for the digital age

In June 2021, a high level EU space event was held in Brussels to mark the adoption of the new EU Space Programme Regulation (EU) 2021/696 and the start of its implementation. The event included a formal signature ceremony for the Financial Framework Partnership Agreement (FFPA) between the Commission, EUSPA and ESA, which is setting out the general obligations to the tasks entrusted to both entities through contribution agreements. In the course of 2021, contribution agreements were also agreed and signed with other entrusted entities, namely EUMETSAT, EEA, MERCATOR OCEAN, ECMWF, FRONTEX and EMSA, to ensure the continuity of Copernicus.

The EU space services provided under EGNOS, Galileo[2] and Copernicus reported excellent performances levels and service availability, in particular with respect to the committed performance levels. In November 2021, improved performance commitments for the Galileo Open Service (OS) and the Public Regulated Service (PRS) were published.

Striving towards technological sovereignty and seeking to increase the number and the quality of services available to the public, work on the development of new innovative services continued, i.e. the development of the Galileo emergency warning service, the high accuracy service, and the authentication services, and improvements to the data management of the Copernicus Data and Information Access Services (DIAS). While the implementation of the Galileo second-generation stage started, intensive discussions were held with stakeholders throughout 2021 on the future of Copernicus and its next

(2) https://www.gsc-europa.eu/electronic-library/galileo-service-performance-reports

generation expansion missions and services, with a focus on green global leadership and on resilient services.

In 2021, the European Court of Auditor's Special Report 07/2021 assessed the effectiveness of the measures taken by the Commission to promote the uptake of services derived from the EU space components (Galileo and Copernicus). The EU auditors found that EU space 'services are launched, but the uptake needs a further boost'. Preparatory activities to address the four recommendations started in 2021.

The number of Copernicus, Galileo and EGNOS users is steadily on the increase, transforming our societies, modernising transport, enabling precision farming and influencing human behaviour in the cities and rural areas to become 'greener' and more 'sustainable'. Nowadays, chipsets processing Galileo signals are widely available and there are already more than 2.5 billion Galileo enabled smartphones in use. It is also positive that the number of registered users downloading Copernicus data and information have increased, reaching a total of 428 600 registered Copernicus users by end of 2021[3].

The first calls under Horizon Europe (space) were published covering research and innovation in areas such as access to space, the future space ecosystem with on-orbit operations and a demonstration mission and quantum technologies. In December 2021, the European Innovation Council (EIC) Horizon Prize for a European Low Cost Space Launch solution was awarded (value of EUR 10 million) to the German company Isar Aerospace. The prize encouraged the development of a European technologically non-dependent solution for launching light satellites into Low-Earth Orbit (LEO) with a committed schedule and orbit.

To support SMEs and start-ups in space, the first action under the CASSINI Space Entrepreneurship Initiative were rolled out with a focus on improving business skills among entrepreneurs and facilitate access to finance for growing companies in 2021-2027.

The organisation of Space dialogues with key international partners could not resume in 2021 due to restrictions linked to the COVID-19 pandemic. Yet, DG DEFIS worked on preparations in view of their organisation when the sanitary situation allows. In September 2021, DG DEFIS signed an administrative arrangement on cooperation on the European Global Navigation Satellite System (EGNSS) with Colombia.

In light of an increasing number of space debris and the exponential increase of space traffic, DG DEFIS was preparing the EU approach to space traffic management[4] in 2021. A call for evidence on the 'EU Strategy for Space Traffic Management' was published via the 'Better Regulation Portal'.

DG DEFIS contributed to the EU's new industrial strategy for Europe (led by DG GROW) adopted on 5 May 2021[5] which is essential for driving forward the green and digital transitions. To implement activities related to the recovery of the aerospace and defence

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([3]) Worldwide users registered on European Copernicus data access portals.

([4]) This initiative is announced under 2022 CWP with adoption expected by Q1 2022.

([5]) COM(2021) 350 final

ecosystem from the COVID-19 crisis, DG DEFIS created a new expert group on policies and programmes relevant to EU space, defence and aeronautics (SDA) industry. DG DEFIS also played a leading role in the preparation of the action plan on synergies between civil, defence and space industries and its implementation. The actions are contributing to strengthen Europe's technological edge and improve the resilience of defence, space, and related civil industries. DG DEFIS was in particular driving forward several key actions to achieve synergies such as those on the Observatory for Critical Technologies, disruptive technologies, innovation and the flagships on space traffic management and on secure connectivity.

A stronger Europe in the world

In June 2021, a high-level event was held in Brussels to mark the adoption of the European Defence Fund Regulation (EU) 2021/697[6] and the start of its implementation. The detailed objectives, activities and budget spending plans for 2021 were set out in the first annual 2021 EDF work programme.

By December 2021, the closure of the first 23 EDF calls for proposals resulted in a total of 142 proposals, among which more than 40% in relation to non-thematic calls dedicated to SMEs and to disruptive technologies for defence. In total, 1 111 different entities, thereof around 50% SMEs, from all the EU Member States (except Malta) and Norway participated in the proposals submitted.

With regard to the precursor programmes of the EDF, under the EDIDP, two award decisions for the EDIDP competitive calls (2020), covering the award of EUR 158.2 million to 26 strategic defence capability development projects and the award decision for EDIDP direct awards (2019 – 2020), were both adopted on 30 June 2021[7]. Under the PADR, three defence disruptive technology for collaborative research projects (METAMASK, PRIVILEGE and SPINAR) started their cutting-edge high-risk research, leading to game-changing impacts in the defence context.

DG DEFIS coordinates the Commission's activities contributing to improving military mobility within Europe. The third progress report on the implementation of the Action Plan on Military Mobility from October 2020 to September 2021[8] (Joint Report DEFIS/ EEAS/MOVE) was released.

DG DEFIS contributed to the Joint Communication on "A stronger EU engagement for a peaceful, sustainable and prosperous Arctic"[9] adopted in October 2021, making sure that EU space-enabled services are in support of the EU Arctic Policy. For example, Copernicus

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(6) Regulation (EU) 2021/697 of the European Parliament and of the Council of 29 April 2021 establishing the European Defence Fund and repealing Regulation (EU) 2018/1092 by co-legislators on 29 April 2021, published on 12 May 2021 (OJ L 170, 12.5.2021, p.149 -177), applicable from 1 January 2021.

(7) C(2021) 4877 and C(2021) 4883

(8) JOIN(2021) 26

(9) JOIN(2021) 27 final

helps to monitor the impact of climate change in the Arctic region, while the Galileo Search and Rescue Service (SAR) is important to help sailors and pilots operating in hostile environments.

EU space diplomacy actions were pursued to strengthen Europe's role as a strong global space actor. In 2021, DG DEFIS started implementing a EUR 6 million 'Global Action on Space' (co-led and financed by the Foreign Partnership Instrument) to promote the EU Space Programme globally, provide information on promising markets and foster new business opportunities for the EU Space ecosystem in over 40 countries around the world.

Promoting our European way of life

DG DEFIS contributed to building Europe's strategic autonomy. Based on the GOVSATCOM component and complementing the EU's satellite navigation (Galileo/EGNOS) and Earth Observation (Copernicus) components of the EU Space Programme, DG DEFIS (jointly with DG CNECT) continued work on a new initiative for an EU space-based global secure communications system[10]. A public consultation was published via the 'Have your say portal' and several workshops to collect feedback from various stakeholders. The feedback received was taken into account in the impact assessment that was carried out to accompany the legislative proposal (adopted in the first quarter of 2022).

The operational EU Space Surveillance and Tracking (SST) component was further reinforced. By the end of 2021, more than 130 European organisations from 23 EU Member States registered to use the EU SST services. More than 260 European satellites benefit from the collision avoidance service.

When it comes to safety and security (dual use) related services, an important milestone was reached in March 2021 when the COSPAS-SARSAT[11] Council authorised the full operational capability of the new functionality of the Galileo Search and Rescue Return Link Service (SAR RLS) clearing the way for worldwide use and the commercialisation of return link service beacons. The Copernicus Emergency Management Service supported partners during major disasters that unfolded throughout the year.

A key EU space policy priority was to pursue an EU autonomous access to space. The Commission undertook intensive negotiations (still ongoing) for the setting up of a framework contract for the supply of aggregated launches services with Arianespace for the EU Space Programme. In addition, preparatory works and consultations with internal and external stakeholders took place in 2021 to develop a common vision of the next generation of EU launch systems, covering the full range of launchers. This work is leading to the set-up of a European Alliance on Space Launchers.

A highlight was the successful Galileo satellite launch in December 2021, bringing the total number of Galileo satellites in orbit to 28 and allowing for increasing the robustness and availability of the existing services.

([10]) Initiatve announced under the CWP 2022 with adoption by Q1 2022.

([11]) http://www.cospas-sarsat.int/en/

Delivering on the Security Union Strategy, DG DEFIS coordinated Commission services in the identification of sectoral hybrid threats resilience baselines at EU level, a first step to track and objectively measure progress. In June 2021, the fifth annual progress report on the implementation of the 2016 Joint Framework on countering hybrid threats and the 2018 Joint Communication on increasing resilience and bolstering capabilities to address hybrid threats[12] was released.

(12) SWD (2021) 729 final

B. Key Performance Indicators (KPIs)[13]

Overall, DG DEFIS progressed well towards reaching its general policy objectives for defence industry and space and in making the EU more secure, sustainable and resilient.

KPI 1 confirms that the overall performance of Galileo and EGNOS progressed well during 2021. The positioning and timing performance of Galileo is better than for any other Global Navigation Satellite System (GNSS) in the world. In addition, it can noted that there were more than 2.5 billion Galileo enabled smartphones in use by 2021. Activities focused on ensuring the continuity of service provision and related operations and in developing new innovative service features i.e. the development of the Galileo emergency warning service, the high accuracy service, and the authentication services which are coming soon.

KPI 2 shows that the number of users of the Copernicus Climate Change Service is growing at a very fast rate and exceeded 100,000 users. This reflects the successful market uptake activities and the growing demand and urgent need for reliable data on climate change. Copernicus, the EU's Earth Observation system is a leading global provider of Earth monitoring data and information with a total number of 428 600 registered Copernicus users for all services by end of 2021[14]. DG DEFIS is building on the achievement of a European autonomous access to environmental knowledge and a major role of the EU at international level.

KPI 3 depicts stable progress in the number of services provided for operational safety and security (including dual-use) related services of the EU Space Programme that are essential for EU security actors. DG DEFIS is in particular exploring synergies between space, security and defence at various levels and this work is resulting in gradual service reinforcements and the introduction of new services as shown in the table.

DG DEFIS rolled out the EDF promoting the further development of a European defence internal market and the competitiveness and innovation capacity of the EU defence industry. In 2021, the first EDF calls for proposals were published on 30 June (11 calls targeting research actions and 12 calls targeting development actions), addressing 37 topics with a total budget of more than EUR 1.2 billion[15]. Given that the EDF is in its early days of implementation, it is not yet possible to report on the EDF KPI results (for more information see PART 1, section C).

(13) Only a selection of 3 out 5 DG DEFIS KPIs are presented here. More information on budget performance of the programmes managed by DG DEFIS can be found on the europa website for the EU Space Programme and the EU Defence Fund.

(14) Worldwide users registered on European Copernicus data access portals.

(15) submission deadline 9 December 2021

1. Number of users of the Copernicus Climate Change Service			
Result indicator: Number of users of the Copernicus Climate Change Service **Source of data:** ECMWF, https://climate.copernicus.eu/			
Baseline (2019)[16]	**Interim Milestone** (2022)	**Target** (2024) The indicated number of users corresponds to 'registered' users of the Copernicus climate change service.	**Latest known results** (2021) The number of users is growing at a very fast rate. Daily up-to-date status can be followed here (end of the page): https://climate.copernicus.eu/
28 000	**70 000**	**80 000**	**100.000**

2. Availability, accuracy, and continuity of services provided by Galileo and EGNOS separately			
Result indicator: Availability, accuracy, and continuity of services provided by Galileo and EGNOS separately[17] **Source of data:** EUSPA (European Union Agency for the Space Programme): EGNOS service provision provider			
Baseline (2020)	**Interim Milestone** (2022)	**Target** (2024)	**Latest known results** (2021)
Galileo availability: 77%	95.5%	Galileo availability: 99.5%	Better than 99.5% at all user locations and for all frequency combinations.
Galileo accuracy: • Horizontal positioning accuracy <= 7.5m (95%) • Vertical positioning accuracy <= 15m (95%)	Galileo accuracy: • Horizontal positioning accuracy <= 5m (95%) • Vertical positioning accuracy <= 8m (95%)	Galileo accuracy: • Horizontal positioning accuracy <= 4m • Vertical positioning accuracy <= 8m	Galileo accuracy: • Horizontal accuracy = 1.8 m (95%) • Vertical accuracy = 2.8 m (95%)
Galileo continuity: Not presently defined	…	Galileo continuity: Will be defined in the applicable issue of the Open Service Service Definition Document (OS SDD)	No figure for continuity has yet been defined
EGNOS availability: • APV-I[18]: Over 99% of the EU territories with more than 99% of availability. • LPV-200[19]: over 90% of the EU territories with more than 99% of	…	EGNOS availability: • APV-I: Over 99% of the EU territories with more than 99% of availability. • LPV-200: over 95% of the EU territories with more than 99%	EGNOS availability: • APV-I: 97% of the EU territories with more than 99% of availability. • LPV-200: over 95% of the EU territories with more than 99% of availability

(16) The Copernicus Climate Change Service started operations in June 2018. The baseline is based on the number of registered users available by 2019.

(17) This indicator is also reported under the Programme Statement for the EU Space Programme, as part of the annual draft general budget of the European Union

(18) APV-I = Approach procedure with vertical guidance, category 1.

(19) LPV-200: Localizer Performance with Vertical guidance to a decision altitude of 200ft

availability		of availability.	
EGNOS accuracy: Horizontal (95%): 2 m, Vertical (95%): 3 m	…	EGNOS accuracy: Horizontal (95%): 1.5 m, Vertical (95%): 2.5 m	EGNOS accuracy: Horizontal (95%): 2 m Vertical (95%): 3 m
EGNOS continuity: • APV-I: over 90% of the EU territories with better continuity than 5*10-4 / 15 sec. • LPV-200: over 80% of the EU territories with better continuity than 5*10-4 / 15 sec.	…	EGNOS continuity: • APV-I: over 95% of the EU territories with better continuity than 5*10-4 / 15 sec. • LPV-200: over 85% of the EU territories with better continuity than 5*10-4 / 15 sec.	EGNOS continuity: • APV-I: over 98% of the EU territories with better continuity than 5*10-4 / 15 sec. • LPV-200: over 93% of the EU territories with better continuity than 5*10-4 / 15 sec.

5. Number of operational safety and security (including dual-use) related services from the EU Space Programme

Result indicator: Number of operational safety and security (including dual-use) related services from the EU Space Programme

Source of data: European Commission (DG DEFIS) - EU Space Programme

Baseline (2020)	**Interim Milestone** (2022)	**Target** (2024)	**Latest known results** (2021)
GALILEO Public Regulated Service (PRS) Initial Service	GALILEO Public Regulated Service (PRS) Initial Service	GALILEO Public Regulated Service (PRS) Full Operational Capability (FOC)[20]	GALILEO Public Regulated Service (PRS) Initial Service
EGNOS Safety of Life (SoL) service	EGNOS Safety of Life (SoL) service	EGNOS Safety of Life (SoL) service	EGNOS Safety of Life (SoL) service
GALILEO Search and Rescue (SAR) service	GALILEO Search and Rescue (SAR) service	GALILEO Search and Rescue (SAR) service	GALILEO Search and Rescue (SAR) service - with return link feature since 2020.
		GALILEO Emergency Warning Service (EWS)	GALILEO Emergency Warning Service (EWS) is in the service definition stage.
Copernicus security service[21]/ emergency service	Copernicus security service/ emergency service	Copernicus security service/ emergency service	Copernicus security service/ emergency service
Space Surveillance and Tracking (SST)	Space Surveillance and Tracking (SST)	Space Surveillance and Tracking (SST)Space Weather (SWE)	Space Surveillance and Tracking (SST)
		Near Earth Objects (NEO)	
		GOVSATCOM (Governmental Satellite Communications)	

(20) The schedule is at risk. New target date is under assessment.

(21) (1) Copernicus Maritime Surveillance Service, (2) Copernicus Border Surveillance Service, (3) Copernicus Support to EU External Actions.

C. Key conclusions on Financial management and Internal control

In line with the Commission's Internal Control Framework DG DEFIS has assessed its internal control systems during the reporting year and has concluded that it is effective and that the components and principles are present and functioning well overall. Some improvements and minor deficiencies were identified which will be taken care of during 2022. These are related to the internal control principles 5, 9 and 13. We refer to the AAR section 2.1.3 for further details.

In addition, DG DEFIS has systematically examined the available control results and indicators, including those for supervising entities to which it has entrusted budget implementation tasks, as well as the observations and recommendations issued by the internal auditors and the European Court of Auditors. These elements have been assessed to determine their impact on management's assurance about the achievement of the control objectives. Please refer to Section 2.1 for further details.

In conclusion, management has reasonable assurance that, overall, suitable controls are in place and working as intended; risks are being appropriately monitored and mitigated and the necessary improvements and reinforcements are being implemented. The Director-General, in his capacity as Authorising Officer by Delegation has signed the Declaration of Assurance.

D. Provision of information to the Commissioner

In the context of the regular meetings during the year between the DG and the Commissioner on management matters, the main elements of this report and assurance declaration, have been brought to the attention of Commissioner Thierry Breton, responsible for the Internal Market.

1. KEY RESULTS and progress towards achieving the Commission’s general objectives and DG's specific objectives

A. European Green Deal

Specific objective 1.1: Reliable data and services of the EU Space Programme are cornerstones for the monitoring of, and transition to climate-neutrality and ecological sustainability

The EU Space Programme and in particular, **Copernicus, the EU Earth Observation system,** provide reliable monitoring data and information on a full, free and open basis, supporting the EU to stay on track towards achieving the EU 2050 climate-neutrality and sustainability objectives. DG DEFIS contributed in 2021 to several **cross-policy legislative initiatives** under the EU Green Deal to ensure the inclusion of EU space-enabled services as enablers of systemic changes towards sustainability and supporting the implementation of:

- the EU biodiversity strategy for 2030; Copernicus-data based solutions help preserving and restoring ecosystems and better understand and protect soil health and ocean health which is paramount for food security and carbon sequestration. The Copernicus climate change service (C3S) that routinely monitors the Earth’s climate, provided throughout 2021 **continuous and reliable information** on key indicators such as temperature, sea ice and CO_2 levels. The number of users of the Copernicus climate change service is steadily increasing since its start in 2018, and has reached almost 100.000 users by end of 2021. The Copernicus Atmosphere Monitoring Service (CAMS) supported COVID-related research into weather, air quality and the spread of the pandemic. The added value of Copernicus CAMS and C3S services in the context of climate adaptation were put at centre stage when presented at the United Nations Climate Change Conference of the Parties (COP26) in November 2021, in Glasgow (UK) during a session called "[Climate adaptation enabled through the Copernicus Services and international cooperation]"[22].

100.000 users Copernicus climate change service

- a sustainable blue economy; DG DEFIS, alongside with DG RTD and DG MARE pushed forward ocean monitoring capabilities, these substantially reinforced with data provided by the recently launched Sentinel 6 Michael Freilich satellite in 2020. The Copernicus Marine Service (CMES) published its 5th Ocean state report and successfully delivered

[22] Session organised by ECMWF in cooperation with the Joint Research Centre (JRC) and DG DEFIS

on eutrophication[23] monitoring at the Exclusive Economic Zone in support of Eurostat reporting of SDG 14.1.

- the EU soil strategy and the ‘Land Use, Land Use Change and Forestry (LULUCF) Regulation: the Copernicus Land Monitoring Service[24] (CLMS) supported monitoring of soil ecosystems and the verification of reported greenhouse gas emissions and carbon dioxide removals from the LULUCF sector which are part of the 2030 Climate and Energy targets.
- the new EU forest strategy and the Commissions’ proposal for a new Regulation to curb EU-driven deforestation; Data and impressive high resolution images were continuously provided with the Copernicus Land Monitoring and Copernicus climate services to monitor and map changes of forest areas and agricultural expansion which is considered one of the main drivers to deforestation.
- the ‘EU Farm to Fork strategy’ with precision farming applications for the sustainability of food systems enabled by the EU Space Programme; DG DEFIS together with DG AGRI proceeded to the second phase of a pilot project for a “Shared Space-and Non-Space Data for Agriculture”. The project tests and promotes data sharing and the scaling up of solutions proposed by the downstream sector and by other EU initiatives (from various DGs, Member States, private actors etc.).
- the EU sustainable and smart mobility strategy; **Galileo, the EU’s satellite navigation system**, provided throughout 2021 high accuracy positioning and navigation signals that are essential for transport solutions. Use of satellite navigation services enables determination of **optimal routes** for cars, public transport, buses or boats by providing accurate positions. Shorter, more efficient journeys result in lowering the amount of time a vehicle engine is engaged. In turn, this allows for significant reduction of the fuel and thus contributes to the reduction of CO2 emissions. For instance, the use of Galileo in road navigation can reduce fuel consumed on the roads in the order of 5 billion litres of fuel[25].

769 EGNOS-based landing procedures in 426 airports

- In air transport, using EGNOS, Europe's regional satellite-based augmentation system, for efficient definition of flight routes permits reduced fuel burn and reduced CO2 emissions. The uptake of EGNOS services showed a further increase in the number of airports with EGNOS landing procedures in 2021, which reached a total of 426 airports.

The EU Space Programme made an important contribution to the **EU taxonomy Regulation** and in defining the qualification of economic activities as environmentally

(23) Harmful algal blooms, dead zones, and fish kills are the results of a process called eutrophication which begins with the increased load of nutrients to estuaries and coastal waters.

(24) and other services/surveys such as LUCAS (Land Use and Coverage Area frame Survey)

(25) European Commission’s Internal Study. “Analysis of the Environmental Impact of the EU Space Programme”, (2020), page 88

sustainable. Technical screening criteria[26] were proposed by DG DEFIS for digital solutions that are exploiting space-based Earth Observations and enabling, among other, climate change mitigation, climate change adaptation, the protection and restoration of biodiversity and ecosystems, pollution prevention and control, sustainable use of waters and marine resources, and their protection.

Preparatory works continued to ensure the contribution of Copernicus to the **Destination Earth[27] project** (part of the Digital Europe Programme and Horizon Europe and co-lead by DG CNECT, DEFIS, RTD and JRC). Destination Earth is developing dynamic, interactive, computing and data intensive "Digital Twins of the Earth", digital multi-dimensional replica of the Earth system. The official launch (first phase)[28] of this project is foreseen for early 2022. It will enable simulations that are expected to speed up the green transition and help plan for major environmental degradation and disasters.

In the area of defence, DG DEFIS proposed funding topics under the European Defence Fund under a **category** dedicated to **"energy resilience & environment transition"**. In 2021, a call for proposals in "energy efficiency and energy management" with a budget of EUR 133 million was published, targeting the areas of energy efficiency for military camps, next generation sources and alternative propulsion.

DG DEFIS also intensified the dialogue with the **civil aeronautics industry**, which was strongly affected by the COVID-19 pandemic and is under pressure to transition to carbon neutrality. Following the EU's industrial strategy, DG DEFIS launched a public consultation addressing the entry into service of hydrogen-powered and fully electric aircraft which emit no CO_2. The results of the consultation confirmed the growing momentum in industry to bring such aircraft to the market and the need for a cross-sectoral initiative to prepare the air transport system for its operation. DG DEFIS intends in 2022 to invite interested actors to join an **Alliance for Zero Emission Aviation** to assess the obstacles to operating hydrogen and electric aircraft and develop solutions to these.

Communication

In the context of the coronavirus pandemic, DG DEFIS has adapted its overall communication strategy towards hybrid/ virtual events and a stronger online presence. One of the key targets has been to communicate the concrete benefits and contribution of the EU Space Programme components to the European Green Deal. DG DEFIS has significantly expanded its communication activities in this regard. In its continuous support to the UN Sustainable Development Goals (SDGs), DG DEFIS has organised a touring exhibition on "Space for Our Planet" and a multimedia platform to showcase real-life stories of how the EU Space Programme contributes to reaching the UN SDGs. The exhibition attracted more than 4.500 participants and was featured at 4 different locations in Brussels and Paris.

(26) Platform on Sustainable Finance - Technical Working Group - Annex: Full list of technical screening criteria August 2021 (europa.eu)

(27) https://ec.europa.eu/digital-single-market/en/destination-earth-destine

(28) 30 months for a value of EUR 150 million

DG DEFIS organisation/presence in environment-related events has been reinforced. Based on the positive experience of last year's activities, DG DEFIS has organised 3 Copernicus workshops with more than 1.000 participants as well as promoting its policies in key international fora such as the Arctic Circle Conference (Iceland) and the COP26 –Glasgow (UK) gathering more than 7.000 participants.

B. Europe fit for the digital age

Specific objective 2.1: Modern and well-functioning EU space-enabled services to support the Union's priorities

A key milestone was achieved with the adoption of Regulation (EU) 2021/696[29] setting out the simplified legal framework of the **new EU Space Programme** and marking the start of its implementation that begun in 2021, after years of intensive negotiations with key stakeholders.

The **first work programmes of the EU Space Programme** were adopted on 18 June[30] by a Commission Decision, setting out the detailed objectives, activities and budget spending plans for 2021 per space component of the programme (including horizontal activities), and seeking to ensure continuity of EU space services provision for EGNOS, Galileo and Copernicus.

The Commission adopted a Commission Implementing Decision in June 2021 authorising the signature of the **Financial Framework Partnership Agreement** (FFPA) between the Commission, EUSPA and ESA, and setting the general obligations applicable to the tasks entrusted to both entities through contribution agreements for the implementation of the EU Space Programme by indirect management[31]. A formal signature ceremony was held on 22 June in Brussels (Belgium). In addition, **contribution agreements** entrusting tasks to other entities were agreed and finalised in 2021 with six contribution agreements[32] signed with EUMETSAT, EEA, MERCATOR OCEAN, ECMWF, FRONTEX and EMSA to ensure the continuity of Copernicus

(29) Regulation (EU) 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme and repealing Regulations (EU) No 912/2010, (EU) No 1285/2013 and (EU) No 377/2014 and Decision No 541/2014/EU, was adopted by co-legislators on 28 April 2021, published on 12 May 2021 in the Official Journal (OJ L 170, 12.5.2021, p. 69–148) and became applicable from 1 January 2021.

(30) C(2021) 4316

(31) C(2021) 4351 and C(2021) 4319

(32) C(2021)5261

Technological sovereignty and EU space service improvements

Galileo showed excellent performances levels and service availability throughout 2021. Continuity of services was assured over the reporting year without interruptions and with a high level of satisfaction from users. In November 2021, **improved performance commitments for the Galileo Open Service (OS)** were published in the service definition document[33]. Work on the development of innovative upgrades and new services continued, seeking to increase the number and the quality of Galileo services available to the public. This includes the Emergency Warning Service, which will provide warnings to users in a geographical area in case of natural disasters or other critical situations; the High Accuracy Service, providing positioning accuracy of 20 cm; and Authentication Services, to ensure the authenticity of signals processed by Galileo receivers, which is needed for road tolling and will enable many other innovative applications.

In 2021, the implementation of the Galileo second generation stage started. On 20 January the Commission awarded **1.47 billion EUR in contracts to launch the 2nd generation of Galileo satellites** to ThalesAleniaSpace (Italy) and Airbus Defence & Space (Germany) following an open competition. The first satellites of this second generation will be placed in orbit by the end of 2024. Thanks to their new capabilities, which are based on cutting edge technologies (digitally configurable antennas, inter-satellite links, propulsion systems 100% electric), these satellites will improve the precision of the Galileo system, its robustness against interference and jamming and the resilience of its signal.

All six **Copernicus services** are operational and were provided in support of atmosphere monitoring, marine environment monitoring, land monitoring, climate change monitoring, emergency management monitoring and security. Copernicus data and information products are made available and accessible to any citizen, and any organisation around the world on a free, full, and open basis. Services continuity was assured over the reporting year without interruptions and with a high level of satisfaction from users, despite some minor difficulties operating remotely due to the COVID-19 crisis. The system demonstrated its robustness and agility.

The Commission (with the technical support of ESA) held intensive discussions with stakeholders for the set-up of **Copernicus 2.0** and its next generation High Priority Candidate Missions (HPCM) throughout 2021, currently known as Copernicus Sentinel expansion missions. A final Commission decision is expected in the course of 2022.

Data management

Work on the evolution of the **Copernicus Data and Information Access Services** (DIAS) continued to further improve data management and ease user access and exploitation

(33) https://www.gsc-europa.eu/sites/default/files/sites/all/files/Galileo-OS-SDD_v1.2.pdf

capacities. A call for tender for the access and exploitation of data generated from the Copernicus Sentinel satellites was launched in December 2021.

DG DEFIS worked closely with JRC and other Commission Services in the set-up of the **Knowledge Centre for Earth Observation (KCEO),** a virtual entity bringing together experts and knowledge from different locations inside and outside the European Commission. On 20 April 2021, the official KCEO launch was announced with an online event, attended by over 700 participants.

During the pandemic, DG DEFIS joined forces with ESA and created the '**Rapid Action Coronavirus Earth Observation' dashboard**, also known as RACE. This platform measures the environmental, social and economic effects of the coronavirus lockdown and monitor post-lockdown recovery. In 2021, the RACE platform was further promoted and improved by developing more indicators to gauge the social, environmental and economic impact of the crisis.

Communication

On 22 June, DG DEFIS organised the online launch event of the EU Space Programme, which over time, will deliver new technologies providing important benefits for policymakers, entrepreneurs and citizens, and lead to the development of new space technologies, data products and services. This makes the EU Space Programme a key component in advancing the EU's digital transformation and resilience.

To ensure coherent communication towards stakeholders and the public, DG DEFIS has further increased coordination and monitoring of communication-related activities by the respective entities entrusted, EUSPA and ESA, and other Copernicus entrusted entities and partners. Regular coordination meetings were organised focusing on improving communication practices, providing guidance and conforming with the new KPI reporting procedures. For the latter DG DEFIS has also provided a dedicated taxonomy and explanatory video.

Specific objective 2.2: EU Space Programme maximises socio-economic benefits

Framework conditions for market uptake

In 2021, the **European Court of Auditor's Special Report 07/2021** assessed the effectiveness of the measures taken by the Commission to promote the uptake of services derived from the EU space components (Galileo and Copernicus). The EU auditors found that EU space 'services are launched, but the uptake needs a further boost'. Preparatory activities to address the four recommendations started in 2021 focusing on the development of a comprehensive action plan for supporting the uptake of Galileo and Copernicus services whilst leveraging all possible synergies between the two systems.

DG DEFIS continued promoting the use of space data, information and services in EU policies and legislation. The objective of the market uptake activities is to support the development of the downstream European space sector and to promote the widest use of the European space data. The market uptake activities are grouped along the downstream market segments.

The activities continued in ensuring further user-uptake of EGNOS and Galileo in the **aeronautical domain,** including unmanned aerial vehicles. In the area of **connected and automated mobility**, DG DEFIS worked closely on the Commission's proposal for an amendment of the **Directive on the framework for the deployment of Intelligent Transport Systems (ITS)**[34]. The provision of secured and reliable timing and positioning services is an essential element of the effective operation of ITS applications and services in all transport modes.

In the area of location-based services, the **Delegated Regulation (EU) 2019/320** requires smartphone compatibility and interoperability with Galileo during emergency calls to the European emergency number 112 and starts applying in March 2022. Therefore, in 2021 DG DEFIS worked on guidelines for compliance with Delegated Regulation (EU) 2019/3201 to support the notified bodies in charge of radio equipment compliance assessment so as to make sure that smartphone manufacturers comply fully with the requirements.

Over **2.5 billion** Galileo-enabled devices

The number of Copernicus, Galileo and EGNOS users is steadily on the increase, transforming our societies, modernising transport, enabling precision farming and influencing human behaviour in the cities and rural areas to become 'greener' and more 'sustainable'. Nowadays, chipsets processing Galileo signals are widely available and there are already more than **2.5 billion Galileo enabled smartphones in use**. It is also positive that the number of registered users downloading Copernicus data and information have increased, reaching a total of 428.600 registered Copernicus users by end of 2021[35].

Special attention was given to activities focusing on the objective of fostering a Union of Equality. On the occasion of the 'EU Diversity Month', a workshop was organised with stakeholders, promoting gender equality as well the EU Diversity charter with in the defence, aeronautics and space sectors. Taking on a long term perspective, DG DEFIS procured in 2021 a pioneering project to set out a demographic categorisation of the workforce employed in the defence, aeronautics and space sector, with a special focus on 'Equality, Diversity & Inclusion'.

Research and innovation – Horizon 2020 and Horizon Europe

Under the Horizon Europe, **the first work programme of the Horizon Europe Cluster 4** covering the years 2021-2022 and the part on space was adopted in August 2021[36], setting out the detailed objectives, activities and budget spending plans. The budget envelope for the 2021-2022 Horizon Europe space work programme is EUR 520 million.

(34) https://eur-lex.europa.eu/legal-content/FR/TXT/?uri=COM:2021:813:FIN
(35) Worldwide users registered on European Copernicus data access portals.
(36) European Commission Decision C(2021)6096 of 23 August 2021.

The first published calls under Horizon Europe (space) covered areas such as access to space, with major efforts devoted to reusable launcher technologies and low cost, high thrust propulsion; Earth Observation technologies; future space ecosystem with on-orbit operations aiming at a demonstration mission by 2026-27; the Copernicus service evolution and Copernicus and Galileo downstream applications. Furthermore, **critical space technologies for European non-dependence** were addressed stemming from ongoing trilateral cooperation between the Commission, EDA and ESA. Horizon Europe also continues to provide research funding for the further development of the EU Space Programme components Galileo/EGNOS, Copernicus, SSA-SST and GOVSATCOM.

Driving forward the development of **quantum technologies**, two specific Horizon Europe call topics were prepared with ESA: The first one related to the development of Quantum Key Distribution (QKD) technologies (embedded in end-to-end satellite communication systems and associated services, EUR 12 million), and the second one related to the technology maturation of Quantum Space Gravimetry (QSG) components (EUR 17 million). In addition, three specific studies were run in 202: Two studies dedicated to the definition of the EuroQCI system in collaboration with DG CNECT which are still ongoing and a third one to derive the critical components of space QKD systems, this last one already completed.

In December 2021, the **European Innovation Council (EIC) Horizon Prize for a European Low Cost Space Launch solution** was awarded (value of EUR 10 million) to the German company Isar Aerospace. The prize encourages the development of a European technologically non-dependent solution for launching light satellites into Low-Earth Orbit (LEO) with a committed schedule and orbit.

Supporting SMEs and start-ups

Boosting innovation and supporting the emerging European New Space industry, the first actions under the **CASSINI Space Entrepreneurship Initiative** were rolled out. The focus was on improving business skills among entrepreneurs and facilitate access to finance for growing companies in 2021-2027. The Commission together with EUSPA organised the first CASSINI Hackathon in June and a second one in November 2021.

Additionally, a **CASSINI In Orbit Demonstration and Validation service** action (EU contribution in 2021 EUR 24 million) was initiated, covering aggregation services for IOD/IOV experiments (if needed), launch services as well as the in-orbit testing of the QKD technology on the test satellite Eagle 1. Procurement and implementation actions are entrusted to ESA. In 2021, procurement actions for the Eagle 1 experiments have been initiated and this work will be continued in 2022.

Global systems with global reach

In the field of Earth Observation, ongoing Copernicus cooperation arrangements with Chile and Brazil provided additional in-situ data for the benefit of Copernicus services and enhanced the quality of their information products. No new arrangement was signed with international partners in 2021 due to restrictions linked to the COVID-19 pandemic.

In the field of satellite navigation, DG DEFIS supported the uptake of Galileo and EGNOS services worldwide and promoted technical cooperation with key international partners. In September 2021, DG DEFIS signed an administrative arrangement on cooperation on the European Global Navigation Satellite System (EGNSS) with Colombia. DG DEFIS also continued to prepare the extension of EGNOS services to the Eastern and Southern Neighbourhoods: negotiations have advanced with Ukraine, and technical consultations took place with Algeria, Libya and Tunisia.

In 2021, DG DEFIS continued promoting the EU Space Programme's operational needs and strategic interests through various international and multilateral fora. The Plenary of the Committee on Earth Observation (CEOS) adopted the roadmap for an international carbon and greenhouse gas monitoring system in support of the Paris Climate Agreement. The European Commission laid the foundation for this cooperation during its term as CEOS chair in 2018. Furthermore, in October 2021 the joint EU-European Free Trade Association (EFTA) Council adopted the Decision to incorporate relevant parts of the EU Space Programme Regulation into the European Economic Area Agreement (for Norway and Iceland).

The organisation of Space dialogues with key international partners could not resume in 2021 due to restrictions linked to the COVID-19 pandemic. Yet, DG DEFIS worked on preparations in view of their organisation when the sanitary situation allows. For example, DG DEFIS organised in November 2021 the 5th edition of the EU-Japan GNSS public-private roundtable with 150 companies participating, ahead of a future EU-Japan Space dialogue.

DG DEFIS also continued its collaboration with DG INTPA to strengthen cooperation on space with countries in Africa. In November, during a Joint Committee under the "Agence pour la sécurité de la navigation aérienne en Afrique et à Madagascar" (ASECNA), the Commission confirmed its support to the upcoming phases for the construction and start of the operation of a Satellite-based Augmentation System (SBAS) in the ASECNA region. Furthermore, the Commission continued to support the work of the Joint Programme Office (JPO), and the 3rd JPO organised its kick-off meeting on 14 December 2021.

Communication

As a pivotal goal of 2021, the network of 'ambassadors' of the EU Space Programme was significantly reinforced. Further synergies were built between the network of Copernicus Relays, the network of the Copernicus Academy and the Galileo Info Centres. These networks act as multipliers on the ground helping to promote the EU Space Programme and the uptake of its data and services all around the world.

Additionally, using the power of professional-grade Copernicus Sentinel satellite imagery, DG DEFIS has developed regular interactions with prime international media outlets, such as Associated Press, Reuters, CNN, Washington Post, Bloomberg, The Times, France24, The New York Times, La Republica, Der Spiegel, or Il Corriere Della Sera, which regularly requested satellite images processed by DG DEFIS for worldwide distribution to better illustrate their news stories.

The first activities under CASSINI - the space entrepreneurship initiative - were promoted over 2021, mainly on social media channels and via a new dedicated sub-webpage on europa.eu.

Upon request of the Slovenian Presidency of the Council of the EU, DG DEFIS organised a one-day online conference named the 'EU Space Day' with a focus on the planned EU space-based secure connectivity initiative.

Other important initiatives

In 2021, under the EDF, specific skills-related defence actions and new EDF calls contributing to the digital transition were rolled-out to support the future innovativeness and **resilience of the defence and aerospace** ecosystem. For example, one call was published on frugal learning for rapid adaptation of artificial intelligence systems and one call on cloud technologies for the development of collaborative services across military domains.

DG DEFIS implemented the **Commission's Pact for Skills** with a partnership (industry, academia, and social partners) committed to upskill and reskill the aerospace and defence sector in the next 10 years. Conferences, workshops and webinars were organised by DG DEFIS encouraging the partnership members to apply to open calls for proposals[37] for collaborative upskilling and reskilling projects.

Under the EDF cyber category aimed to **increase cyber resilience** and with a budget value of EUR 33.5 million, a research call on "Improving cyber defence and incident management with artificial intelligence" and a development call for "Improved efficiency of cyber trainings and exercises" were launched. Ongoing projects under the precursor EDF programmes are additionally contributing to cyber defence capabilities, in particular the EDIDP project ESSOR[38] and the two projects CYBERDE and DISCRETION selected following a SME-dedicated call for innovative solutions. Under PADR, a project on future disruptive technologies in the field of artificial intelligence (AI) was funded, specifically for the detection of explosive devices.

DEFIS participated in the preparation of new legislation on the growing ecosystem of **drone related industries and services** led by DG MOVE. It supports the implementation of Regulation (EU) 2019/945, organising, in particular, the administrative cooperation between market surveillance authorities (ADCO drones) and supervising the development of the related harmonised standards. It contributed to the preparation of the Drone 2.0 strategy to be adopted in 2022. Focus was given in particular to the spin-off and spin-in effects of the customer driven market demands and defence and security requirements. The same technological development can serve a large scope of different sectors with different use-cases for the same technological solutions.

(37) under the European Digital Programme and Erasmus+
(38) European Secure Software-defined Radio (ESSOR)

Reinforcing industrial Strategic Autonomy and in light of an increasing number of space debris and the exponential increase of space traffic, DG DEFIS was preparing the **EU approach to space traffic management**[39] in 2021. Space is increasingly congested, threatening the viability and security of space infrastructure and operations. A call for evidence on the '**EU Strategy for Space Traffic Management'** was run from 22 October until 19 November 2021, published via the 'Better Regulation Portal'.

DG DEFIS contributed to the EU's new industrial strategy for Europe (led by DG GROW) adopted on 5 May 2021[40] which is essential for driving forward the green and digital transitions. A new policy approach was set in motion focusing on increasing resilience and on **strengthening EU's open strategic autonomy**. The Aerospace and Defence (ASD) ecosystem, being among 14 industrial ecosystems put forward by the strategy, is central to this approach and will contribute to achieving these objectives, with progress to be monitored on an annual basis.

To implement activities related to the **recovery of the ASD ecosystem from the COVID-19 crisis**, DG DEFIS created a new expert group on policies and programmes relevant to EU space, defence and aeronautics (SDA) industry. Several virtual workshops with the expert group were organised in 2021, providing valuable inputs to Commission's work on mapping strategic dependencies, and critical technologies for space, defence and civil-related applications.

To strengthen Europe's technological edge and improve the resilience of existing industries, the Commission adopted an 11-point **action plan on synergies between civil, defence and space industries** on 22 February 2021[41]. DG DEFIS plays a leading role in many of the actions, in particular on the flagships STM and Secure connectivity, and in actions related to SMEs', the Observatory for Critical Technologies, standardisation, defence innovation networks and promoting opportunities across programmes and funding opportunities for defence industry entities.

DEFIS kicked-off the work on the **EU Observatory for Critical Technologies (OCT)** (co-led by DEFIS and JRC) by setting up an internal OCT development team working in three subgroups: security; methodology, use cases & requirements; stakeholder engagement. The main focus in 2021 was on the collection of business needs from other DGs, on the development of a security policy and on the methodologies for the data analysis and mapping as well as on the strategy for an efficient engagement with different stakeholders. A first classified report will be prepared by the Commission for selected critical technologies by the end of 2022.

DG DEFIS launched in the fourth quarter of 2021 a request for call for service to obtain a precise snapshot on semiconductors' needs for the defence and space sectors in the

(39) This initiative is announced under 2022 CWP with adoption expected by Q1 2022.
(40) COM(2021) 350 final
(41) COM(2021) 70 final

European Union in the short, medium and longer term. The start of the study is expected in the first quarter of 2022.

2021 has been the first full year of the screening of **Foreign Direct Investments.** DEFIS prioritised and contributed to the assessment of more than 400 cases (as of 23 November 2021) that were notified under first year of implementation or Regulation (EU) 2019/452.

C. A stronger Europe in the world

Specific objective 4.1: Fostered innovation capacity and competitiveness of the European defence industry and strengthened EU defence supply chains due to increased cross-border R&D cooperation involving in particular SMEs and mid-caps

A highlight of 2021 was the adoption of the Regulation (EU) 2021/697[42] establishing the **European Defence Fund (EDF)** and allowing the start of its implementation in 2021. The first **annual 2021 EDF work programme** was adopted on 30 June[43] setting out the detailed objectives, activities and budget spending plans for 2021. Also, **the first part of the 2022 annual work programme** was adopted to complement the 2021 budget envelope to kick start ambitious EDF projects related to large-scale and complex defence capabilities. The first EDF calls for proposals were published on 30 June (11 calls targeting research actions and 12 calls targeting development actions), addressing 37 topics and with a total budget of EUR 1.2 billion.

Following the closure of the 23 EDF calls for proposals on 9 December, 142 proposals had been submitted, among which more than 40% in relation to non-thematic calls dedicated to SMEs and to disruptive technologies for defence. In total, 1 111 different entities, thereof around 50% SMEs, from all the EU Member States (except Malta) and Norway are participating in the proposals submitted.

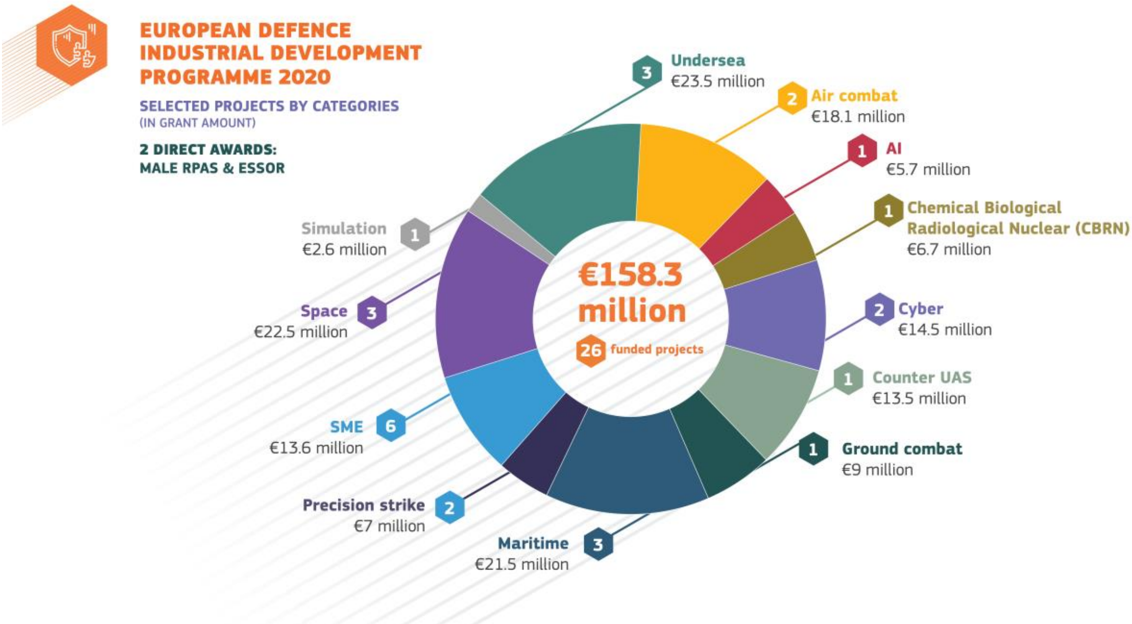
For the preparation of the EDF 2021 calls for proposals, improved submission forms and guidance for applicants were released, taking into account the lessons learned from PADR and EDIDP and the specific additional new features of the EDF. The online submission process for proposals was improved with a first step towards the full integration of the EDF on the corporate eGrants application completed in 2021. Two award decisions for the EDIDP competitive calls (2020) were adopted on 30 June[44], covering the award of EUR 158.2 million to 26 strategic defence capability development projects. All grant agreements have been successfully signed and projects started in 2021.

([42]) Regulation (EU) 2021/697 of the European Parliament and of the Council of 29 April 2021 establishing the European Defence Fund and repealing Regulation (EU) 2018/1092 by co-legislators on 29 April 2021, published on 12 May 2021 (OJ L 170, 12.5.2021, p.149 -177), applicable from 1 January 2021.

([43]) C(2021)4910

([44]) C(2021) 4877 and C(2021) 4883

In addition, an award decision for EDIDP direct awards (2019 – 2020), was adopted on 30 June 2021. Contribution agreements with the Organisation for Joint Armament Cooperation (OCCAR) for the management and implementation of two EDIDP projects were signed in November 2020. Notably, the European Secure Software defined Radio (ESSOR) and European Medium-altitude Long Endurance Remotely Piloted Aircraft System (MALE RPAS) with grant agreements signed by OCCAR with the respective consortium in July 2021. The ESSOR grant agreement entered into force in November 2021 whereas the MALE RPAS grant agreement is expected to enter into force in the first half of the year 2022.[45]



Three disruptive technology projects METAMASK, PRIVILEGE and SPINAR funded under PADR started their cutting-edge, high-risk research, leading to game-changing impacts in the defence context. These projects target a novel and ambitious science-to-technology breakthrough and are driven by young academic researchers, visionary research centres of big companies and research and technology organisations, and as such contribute to the Commission's goal to stimulate innovative disruption in the European defence industry.

The time from submission of complete proposals to informing all applicants of the outcome of the evaluation of their application was seven months, exceeding the period required by Article 194.2 (a) of the Financial Regulation by one month. This deviation was due to the complexity of the programme and the limited resources, both human and technical (such as IT resources) available. Some of the grants were signed six months from the date of informing applicants that they have been successful, instead of the three months required by Article (b) from the Financial Regulation. For those grants, the signature was delayed due to the time necessary to receive the completed 'Programme Security Instructions' from the National Security Authorities and the complexity of the corporate and financial arrangements for the consortium. The application of specific rules linked to double

([45]) This will occur after the linked procurement contract (i.e. the other source of co-financing from the Member States) will enter into force.

comitology[46] is adding to the overall complexity and the delays of the implementation process.

With respect to legislative tools, the Commission continued its constructive dialogue with Member States to ensure the proper implementation of Directive 2009/43/EC on the transfer of defence-related products along with initiating a dedicated discussion, with Member States and industrial actors, on how the tools of this Directive could be utilised to facilitate the implementation of projects funded by the EDF. On 5 March 2021, the Commission adopted Commission Delegated Directive (EU) 2021/1047[47] updating the Annex to Directive 2009/43/EC in line with the updated Common Military List of the European Union. In November 2021, the Commission sent letters of formal notice to eight Member States who had not communicated transposition of the Directive by the relevant deadline.

With respect to Directive 2009/81/EC on defence and security procurement, the Commission continued to monitor Member States' defence procurement programmes and is engaging with Member States to ensure the proper implementation of the Directive. On 10 November 2021, the Commission also adopted Commission Delegated Regulation (EU) 2021/1950[48] updating the thresholds for the application of the Defence and Security Procurement Directive.

Communication

On 30 June, DG DEFIS organised the online launch event of the EDF supporting the competitiveness and innovation capacity of the EU defence industry. To this end, a stronger online presence for defence-related matters was created focusing on increasing awareness of the achievements of the precursor programme, PADR and EDIDP. In addition, a Network of European Defence Fund National Focal Points (NFP) was set up to support the EDF implementation.

A digital Handbook for SMEs on the Directive for the **transfer of defence-related products** was developed, translated in all EU official languages and made available on the Commission's Internet site[49]. It provides a summary of the benefits that this Directive can give to SMEs active in the defence sector, along with concrete recommendations.

In terms of events, DG DEFIS organised a series of EDF Info Days and thematic conferences in different Member States, and presented its policies at international fairs and exhibitions all targeting to raise visibility and awareness about the EDF among SMEs and other stakeholders.

(46) The double comitology is a decision-making principle embedded in the EDF regulation. The EDF Programme Committee gives an opinion on the EDF annual work programme to be adopted by the Commission as well as to the selection of projects to be funded following the EDF calls for proposals.

(47) OJ L 225, 25.6.2021, p. 69-101.

(48) OJ L 398, 11.11.2021, p. 19-20.

(49) https://ec.europa.eu/defence-industry-space/download-defence-transfers-directive-handbook-smes_en

Other important initiatives

In 2021 the Commission explored the possibility of **accession of the EU to relevant UN Treaties and Conventions.** In this regard, DG DEFIS has been working on different steps leading to the participation of the EU to the Agreement on the rescue of astronauts, the return of astronauts and the return of objects launched into outer space (the Rescue Agreement), to the Convention on International Liability for Damage Caused by Space Objects (the Liability Convention), and to the Convention on the Registration of Objects Launched into Outer Space (The Registration Convention).

DG DEFIS aimed in 2021 to reinforce the role that **EU space-enabled services** can play **in support of the EU Arctic Policy.** DEFIS contributed to the new Joint Communication from the Commission and the High Representative to the European Parliament and the Council on "a stronger EU engagement for a peaceful, sustainable and prosperous Arctic"[50] adopted in October 2021. Copernicus helps to monitor the impact of climate change in the Arctic region and provides unrestricted and free data that can help scientists better understand the phenomenon. The Galileo Open Service can provide very accurate positioning and timing data for users in the Arctic region, while the Galileo Search and Rescue Service (SAR) is important to help sailors and pilots operating in hostile environments. Finally, the EGNOS Safety of Life Service is available in certain areas in the Arctic and enables all airports in mainland Norway, Sweden and Finland to use EGNOS for landing aircraft.

Regarding space diplomacy, DG DEFIS worked to support and foster opportunities for EU companies in foreign markets, thus strengthening Europe's role as a strong global space actor. In 2021, **DG DEFIS started implementing a EUR 6 million 'Global Action on Space'** (co-led and financed by the Foreign Partnership Instrument) to promote the EU Space Programme globally, provide information on promising markets and foster new business opportunities for the EU Space ecosystem in over 40 countries around the world. Furthermore, the European Commission organised events to raise awareness of the EU Space Programme worldwide. For instance, the Commission organised a Technical Assistance and Information Exchange (TAIEX) workshop on EU space data applications for Uzbekistan, as well as Galileo hackathons for ASEAN countries via the Horizon 2020 programme.

DG DEFIS coordinates the Commission's activities contributing to improving **military mobility** within Europe. In September 2021, the Commission and the High Representative of the EU for foreign affairs and security policy jointly issued the third progress report on the implementation of the Action Plan on Military Mobility from October 2020 to September 2021[51]. In the work programme 2021, the European Defence Fund included a topic on the development of a digital system for secure and quick exchange of information related to Military Mobility with a dedicated budget of EUR 50 million.

([50]) JOIN(2021) 27 final

([51]) JOIN(2021) 26

D. Promoting our European way of life

Specific objective 5.1: Security actors have access to EU autonomous tools, space-enabled services, and technologies, needed to build resilience to security threats, safety hazards and crisis situations

Secure communications and cyber security

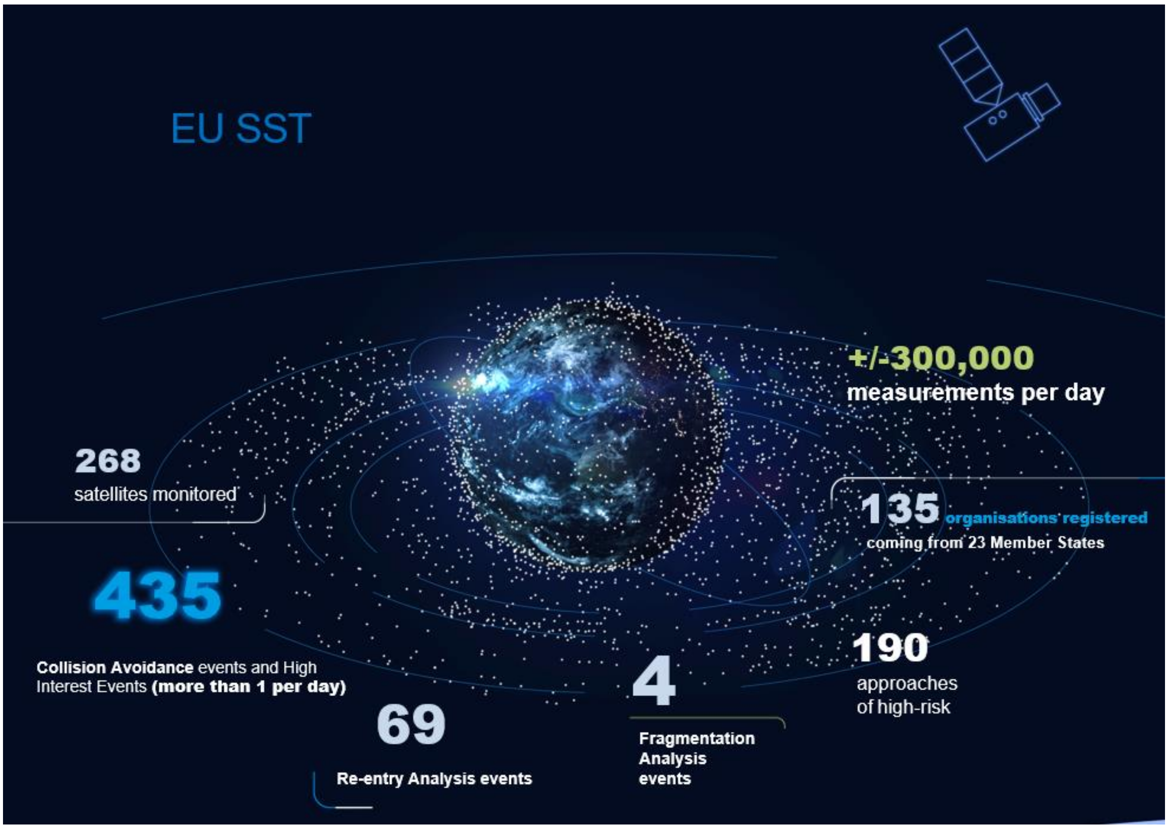
DG DEFIS contributes to building Europe's strategic autonomy. Based on the GOVSATCOM component and complementing the EU's satellite navigation (Galileo/EGNOS) and Earth Observation (Copernicus) components of the EU Space Programme, DG DEFIS (jointly with DG CNECT) continued work on **a new initiative for an EU space-based global secure communications system**[52]. This work involved several activities:

- to explore technical aspects and the potential service provision models of this initiative, the Commission launched an **initial system study** in December 2020 to suggest specifications for the mission, architecture, frequencies, cost, governance and business model. Major actors in the field have teamed-up in a consortium to deliver this system study: large system integrators, launch service providers, satellite networks operators, services providers and telecom providers. This consolidation study will be completed in February 2022.
- to support **European New Space's** dynamic involvement in the enlarged space industrial ecosystem valorising their innovative potential with cutting edge differentiators, the Commission launched a dedicated study. Following an open call for tenders, two consortia were awarded a contract comprising start-ups and several SMEs.
- to consult key stakeholders, the Commission elaborated an **inception impact assessment** and launched a **public consultation** via 'Have your say portal' that was open for feedback until 23 September 2021. The feedback received was taken into account in the **impact assessment** that was carried out to accompany the **legislative proposal** which is expected to be adopted in the first quarter of 2022. In addition, the Commission organised workshops to collect further feedback from various stakeholders.

The added value of the operational **EU Space Surveillance and Tracking (SST**) capacity and soon the reinforced **Space Situational Awareness (SSA)**[53] component under the new EU Space Programme, has been demonstrated in 2021. Under the fragmentation service, the EU SST confirmed the detection and monitored space debris from destruction of a satellite in low orbit (COSMOS 1408) following an anti-satellite test conducted by Russia on 15 November 2021. Around 900 space debris were detected which put at risk crewed and un-crewed space activities. This event clearly demonstrated that space is increasingly contested and must become a fully-fledged dimension of a future works on a European

(52) Initiatve announced under the CWP 2022 with adoption by Q1 2022.
(53) The SSA component will cover EU SST, Space Weather and Near Earth Objects (NEO).

space and defence strategy[54]. To further the development of SSA capabilities to detect, track, identify and characterise threats against space assets, three grants were awarded in 2021 under EDIDP[55] and future ones are prepared under EDF.



The grant agreement with the **current EU SST consortium** was extended until 31 December 2022 to ensure continuity of EU SST services during the set-up of the new EU SST partnership. A Commission Implementing Act detailing the procedure of the establishment of the **future EU SST Partnership** was prepared and discussed in detail with the Member States, expected to be adopted in the first quarter of 2022.

When it comes to **personal safety**, the Galileo's Search and Rescue service reduces drastically the time to detect emergency distress beacons from up to three hours to just ten minutes. In March 2021, an important milestone was reached when the COSPAS-SARSAT[56] Council authorised the full operational capability of the new functionality of the **Galileo Search and Rescue Return Link Service (SAR RLS)**, clearing the way for

(54) https://ec.europa.eu/commission/commissioners/2019-2024/breton/announcements/statement-thierry-breton-european-commissioner-space-following-russian-test-anti-satellite-weapon_en

(55) (1) 'Sensors for Advanced Usage & Reconnaissance of Outerspace situation (SAURON
(2) 'Innovative and Interoperable Technologies for Space Global Recognition and Alert (INTEGRAL) covering Command and Control' and (3) 'Multi-national Development Initiative for a Space-based missile early-warning architecture (ODIN's EYE)'.

(56) http://www.cospas-sarsat.int/en/

worldwide use and the commercialisation of return link service beacons. This service where an automatic acknowledgement message is provided back to the user informing them that their request for help has been received, is uniquely provided by Galileo.

Galileo is introducing an innovative **Emergency Warning service** in its portfolio of services which is relevant for security actors. In 2021, works focused on the definition of this service with Member States civil protection entities, leading to a service baseline ready to be assessed with EUSPA and ESA.

The **Copernicus emergency service** continued supporting actions of the European Union, at local, national and international level, during major natural disasters and distress at sea. To mention some examples: support to Germany, Belgium, Netherlands and Luxembourg during the extreme floods in July 2021, support to Greece when faced with a devastating fire season in July/August 2021, and support to Spain to manage the volcanic eruption in September/October 2021 in La Palma, Canary islands of Spain. In 2021, 58 different Member States administrations, EU institutions and international organizations continued using the Copernicus Maritime surveillance service to support the monitoring of activities at sea that have an impact on areas such as: fisheries control maritime safety and security law enforcement marine environment (pollution monitoring) and other operations linked to anti-piracy and defence.

Access to space and satellite launches in 2021

EU autonomous access to space is a key priority and at the heart of EU's space policy. The Commission undertook intensive negotiations for a set-up of a framework contract for the supply of aggregated launches services to Arianespace for all EU Space Programme.

DG DEFIS also intensified the dialogue with ESA, Member States, national space agencies and EU launcher industries to jointly define a common roadmap for the next generation of launchers and technologies, based on innovative, competitive, resilient and sustainable EU launcher value chains.

Preparatory works and consultations with internal and external stakeholders took place in 2021 to develop a common vision of the next generation of EU launch systems, covering the full range of launchers. This work will lead to the set-up of **a European Alliance on Space Launchers,** as well as the 'Flight ticket' to stimulate new European space transportation services, including micro launchers, through public sector support, both expected to be launched in 2022. It will also lead to the definition of new actions to improve access to test and launch facilities for new EU space transportation solutions.

A highlight was the **Galileo satellite launch (#L11).** On 5 December, two new Galileo satellites were successfully launched from the European Spaceport in Kourou, French Guiana, bringing the total number of Galileo satellites in orbit to 28. This allows for increasing the robustness and ensuring the availability of the existing services. The next Galileo satellite launch is already planned to take place in April 2022.

The Copernicus constellation counted 8 satellites[57] in orbit by the end of 2021. The Copernicus Sentinel 6 Michael Freilich launched in 2020 was fully commissioned and delivers full open and free data, information and services as planned. Throughout 2021, discussions continued on the next generation satellites, scope and schedule, underpinning the future evolution for Copernicus.

Communication

On 27 September 2021, DG DEFIS organised the "Galileo SAR meet" with a real-time demonstration event and a back-to-back symposium in partnership with the Belgian authorities, 40th Squadron of Belgian Air Force base in Koksijde (Belgium). The event was seeking to reinforce existing partnerships with the national and regional Search and Rescue (SAR) authorities and collect feedback for the evolution of the Galileo SAR service. It was widely attended and attracted broad media coverage.

Another highlight was the 11th launch of Galileo satellites, where two new satellites joined the Galileo constellation and with broad media coverage that resulted.

Other important initiatives

DG DEFIS, working closely with EEAS, coordinated the implementation of the initiatives laid down in the EU security strategy. These include for example the coordination of the mainstreaming of the hybrid considerations into policy making, of the creation of a restricted online platform for Member States' reference on counter-hybrid tools and improving situational resilience. DG DEFIS coordinated the Commission departments to identify **sectoral hybrid threats resilience baselines at EU level.** This constitutes the first step to systematically track and objectively measure progress in preventing and protecting Member States and EU institutions and bodies against hybrid threats. This work complements the comprehensive mapping of existing hybrid threats related measures and a list of corresponding policy documents at EU level that was published in June 2020. In the second semester 2021, DG DEFIS, in cooperation with the EEAS, **supported the Member States in elaborating the future EU hybrid toolbox**. In June 2021, **the fifth annual progress report** on the implementation of the 2016 Joint Framework **on countering hybrid threats** and the 2018 Joint Communication on increasing resilience and bolstering capabilities to address hybrid threats[58] was released.

([57]) Sentinel-1A, 1B, 2A, 2B, 3A, 3B, 5P, 6A
([58]) SWD (2021) 729 final

2. MODERN AND EFFICIENT ADMINISTRATION AND INTERNAL CONTROL

2.1. Financial management and internal control

Assurance is provided on the basis of an objective examination of evidence of the effectiveness of risk management, control and governance processes. This examination is carried out by management, who monitors the functioning of the internal control systems on a continuous basis, and by internal and external auditors. The results are explicitly documented and reported to the Director-General. The following reports have been considered:

- the reports from Authorising Officers by Delegation in other DGs managing budget appropriations in cross-delegation;
- the results of the DG's supervisory controls on the operational and financial reporting of these bodies;
- the contribution by the Head of Unit in charge of Risk Management and Internal Control, including the results of internal control monitoring at DG DEFIS;
- the recorded exceptions, non-compliance events and any cases of 'confirmation of instructions' (Art 92.3 FR);
- the results of ex-post controls;
- the limited conclusion of the Internal Auditor on the state of internal control, and the observations and recommendations reported by the Internal Audit Service (IAS);
- the observations and the recommendations reported by the European Court of Auditors (ECA).

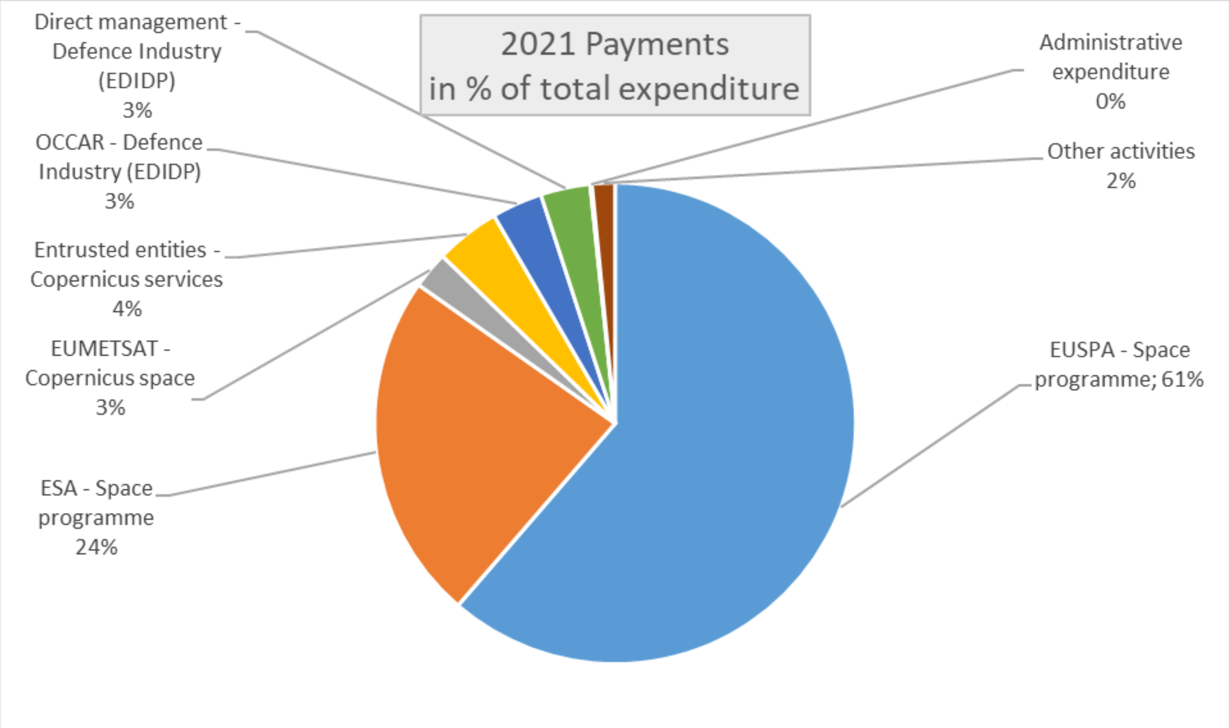
These reports result from a systematic analysis of the available evidence. This approach provides sufficient guarantees as to the completeness and reliability of the information reported and results in a complete coverage of the budget delegated to the Director-General of DG DEFIS.

2.1.1. Control results

This section reports and assesses the elements identified by management which support the assurance on the achievement of the internal control objectives[59]. The DG's assurance building and materiality criteria are outlined in AAR Annex 5. Annex 6 outlines the main risks together with the control processes to mitigate them and the indicators used to measure the performance of the relevant control systems. DG DEFIS transactions are carried out under both direct and indirect management modes. The following chart gives an overview of the types of expenditure with their respective percentage in the total payments

(59) 1) Effectiveness, efficiency and economy of operations; 2) reliability of reporting; 3) safeguarding of assets and information; 4) prevention, detection, correction and follow-up of fraud and irregularities; and 5) adequate management of the risks relating to the legality and regularity of the underlying transactions, taking into account the multiannual character of programmes as well as the nature of the payments (FR Art 36.2). The 2nd and/or 3rd Internal Control Objective(s) (ICO) only when applicable, given the DG's activities.

made in 2021; the total amount paid in 2021 amounts to 2.927.884.583 EUR as shown in the table below.



The largest part of DG DEFIS expenditure was implemented in indirect management through contribution agreements (especially for Space activities). In 2021, 95.17% of the expenditure was implemented in indirect management, 4.71% in direct management and 0.12% for administrative expenses.

Activity		Entity / Subsidy	Payments 2020	% on total payments	Main ICO Indicator
Direct Management	**Grants**	DEFIS	105.958.957	3,62%	Time to pay; Time to grant; Overall cost of Control; No
	Procurement	DEFIS	22.044.096	0,75%	
	Prizes	DEFIS	10.000.000	0,34%	
Indirect Management	**Delegation Agreements/Contribution Agreements**	EUSPA	1.795.832.532	61,34%	CA objectives achieved; Time to pay; Overall cost of control; No Olaf Cases; Clean opinion on accounts; Detected or estimated error rate
		ESA	686.115.567	23,43%	
		EUMETSAT	72.665.000	2,48%	
		ECMWF	62.161.521	2,12%	
		MERCATOR	26.301.000	0,90%	
		EEA	14.931.000	0,51%	
		FRONTEX	9.922.000	0,34%	
		EMSA	8.200.000	0,28%	
		SATCEN	5.664.000	0,19%	
		EDA	4.836.200	0,17%	
		OCCAR	99.709.886	3,41%	
Administrative Expenses		DEFIS	3.542.825	0,12%	
Total			**2.927.884.583**	**100%**	

DG DEFIS does not have to disclose any case of financing not linked to costs (Financial Regulation (FR) art 125.3[60]), nor cases of derogations from the principle of non-retroactivity

([60]) The authorising officer responsible shall report on financing not linked to costs pursuant to points (a) and (f) of the first subparagraph of paragraph 1 of this Article in the annual activity report referred to in Article 74(9).

principle for grants (FR art 193.2). The cases of 'confirmation of instructions' (FR art 92.3[61]) are listed in the DG DEFIS register of exceptions and non-compliance events. DG DEFIS has signed a Financial Framework Partnership Agreement >4 years with EUSPA and ESA (FR art 130.4). DG DEFIS has to disclose cases of flat rates >7% for indirect costs decided by Commission Decisions (FR art 181.6[62]) as PADR, EDIDP and EDF finance indirect costs at flat rate 25%. This is included in the financing decision (for PADR) and regulations.

1. *Effectiveness of controls*

a. Legality and regularity of the transactions

DG DEFIS is using internal control processes to ensure the adequate management of the risks relating to the legality and regularity of the underlying transactions it is responsible for, taking into account the multiannual character of programmes and the nature of the payments concerned. The control objective is to ensure that the residual error rate or the risk of error does not exceed 2%.

DG DEFIS's portfolio consists of different segments all with relatively low error rates. The relatively higher level of error in Horizon 2020 grants is linked to the reimbursement of eligible costs and errors in particular on personnel costs mainly due to the beneficiaries' lack of thorough understanding of the rules. New entrants and Small and Medium Enterprises (SMEs) are more prone to this type of errors. However, since Horizon 2020 is a multi-annual programme, the representative detected error rate of 2.29 % should be considered within a time perspective. Due to the cleaning effect of audits the cumulative residual error rate is only 1.60 % (for more detailed information see Annex 5). The simplifications introduced under Horizon Europe should reduce the number of errors made by the beneficiaries and lead to a further decrease in the error rate in the long run.

For the 2021 reporting year, no serious control issues were signalled by the operational units regarding the work implemented by entrusted entities (95% of the budget). Through the monitoring and supervision work, which included regular contacts and analysis of periodic reports as well as audit reports, no legality and regularity risks were identified.

([61]) An authorising officer by delegation or sub-delegation who receives a binding instruction which he or she considers to be irregular or contrary to the principle of sound financial management, in particular because the instruction cannot be carried out with the resources allocated to him or her, shall inform the authority from which he or she received the delegation or subdelegation about that fact in writing. If the instruction is confirmed in writing and that confirmation is received in good time and is sufficiently clear, in that it refers explicitly to the points which the authorising officer by delegation or subdelegation has challenged, the authorising officer by delegation or subdelegation shall not be held liable. He or she shall carry out the instruction, unless it is manifestly illegal or constitutes a breach of the relevant safety standards. The same procedure shall apply in cases where an authorising officer considers that a decision, which is his or her responsibility to take, is irregular or contrary to the principle of sound financial management or where an authorising officer learns, in the course of acting on a binding instruction, that the circumstances of the case could give rise to such a situation. Any instructions confirmed in the circumstances referred to in this paragraph shall be recorded by the authorising officer by delegation responsible and mentioned in his or her annual activity report.

([62]) The authorising officer responsible may authorise or impose, in the form of flat-rates, funding of the beneficiary's indirect costs up to a maximum of 7 % of total eligible direct costs for the action. A higher flat rate may be authorised by a reasoned Commission decision. The authorising officer responsible shall report in the annual activity report referred to in Article 74(9) on any such decision taken, the flat rate authorised and the reasons leading to that decision.

DG DEFIS's relevant expenditure, estimated overall risk at payment, estimated future corrections and risk at closure are disclosed in the Table X ("Estimated risk at payment and at closure") provided below.

The estimated overall risk at payment for 2021 expenditure amounts to EUR 9.81 million, representing 0.62 % of the DG's total relevant expenditure for 2021. This is the Authorising Officer by Delegation's (AOD) best, conservative estimation of the amount of relevant expenditure not in conformity with the contractual and regulatory provisions applicable at the time the payment was made for 2021. This expenditure will subsequently be subject to ex-post controls and a proportion of the underlying errors will be detected and corrected in subsequent years. The conservatively estimated future corrections for 2021 expenditure amount to EUR 0.17 million. The difference gives an estimated overall risk at closure of EUR 9.64 million, corresponding to 0.61 % of the DG's total relevant expenditure for 2021.

In the context of the protection of the EU budget and to have a view at Commission level, the DGs' estimated overall risk at payment, estimated future corrections and risk at closure are consolidated at Commission level in the Annual Management Performance report (AMPR).

Table X: Estimated risk at payment and at closure (amounts in EUR million)

The full and detailed version of the table is provided in Annex 9.

DEFIS	2021 Management Mode (1)	2021 Relevant Expenditure (in €) (2) - (3) + (4) + (5) - (6) = (7)	2021 Estimated overall risk at PAYMENT (in €) (7) x (8) = (9)		Estimated future corrections (in €) (7) x (10) = (11)		2021 Estimated overall risk at CLOSURE (in €) (9) - (11) = (12)	
ALL BUDGETS : Operational and Administrative expenditure (including E			Low	High	Low	High	Low	High
Administrative credits		**3.542.824,88 €**	17.714,12 €	17.714,12 €	2.479,98 €	2.479,98 €	€ 15.234,15	€ 15.234,15
Heading 1 -Own Procurement		**23.406.380,95 €**	117.031,90 €	117.031,90 €	16.384,47 €	16.384,47 €	€ 100.647,44	€ 100.647,44
Heading 1 -Prize		**10.000.000,00 €**	50.000,00 €	50.000,00 €	7.000,00 €	7.000,00 €	€ 43.000,00	€ 43.000,00
Heading 5 -Grants - EDIDP		**- €**	- €	- €	- €	- €	€ -	€ -
Heading 1 -Grants - Others (1)		**7.712.627,27 €**	38.563,14 €	38.563,14 €	5.398,84 €	5.398,84 €	€ 33.164,30	€ 33.164,30
Heading 1 -International organisations - ESA Copernicus		**340.095.335,71 €**	1.700.476,68 €	1.700.476,68 €	- €	- €	€ 1.700.476,68	€ 1.700.476,68
Heading 1 -International organisations - ESA IOV/IOD		**11.100.000,00 €**	55.500,00 €	55.500,00 €	- €	- €	€ 55.500,00	€ 55.500,00
Heading 1 -International organisations - ESA GNSS Galileo		**151.279.719,34 €**	756.398,60 €	756.398,60 €	- €	- €	€ 756.398,60	€ 756.398,60
Heading 1 -International organisations - ESA Govsatcom/SSA		**- €**	- €	- €	- €	- €	€ -	€ -
Heading 1 -International organisations - ESA Horizon 2020/Europe		**35.091.329,36 €**	175.456,65 €	175.456,65 €	- €	- €	€ 175.456,65	€ 175.456,65
Heading 1 -International organisations - MERCATOR (Copernicus)		**25.777.849,38 €**	128.889,25 €	128.889,25 €	- €	- €	€ 128.889,25	€ 128.889,25
Heading 1 -International organisations - ECMWF (Copernicus)		**59.115.215,55 €**	295.576,08 €	295.576,08 €	- €	- €	€ 295.576,08	€ 295.576,08
Heading 1 -International organisations - EUMETSAT (Copernicus)		**51.205.487,89 €**	256.027,44 €	256.027,44 €	- €	- €	€ 256.027,44	€ 256.027,44
Heading 5 -International organisations - OCCAR		**- €**	- €	- €	- €	- €	€ -	€ -
Heading 1 -Agencies - EUSPA Subsidy		**35.449.479,40 €**	177.247,40 €	177.247,40 €	- €	- €	€ 177.247,40	€ 177.247,40
Heading 1 -Agencies - EUSPA EGNOS		**163.755.460,50 €**	2.112.445,44 €	2.112.445,44 €	- €	- €	€ 2.112.445,44	€ 2.112.445,44
Heading 1 -Agencies - EUSPA GALILEO		**592.454.954,49 €**	2.962.274,77 €	2.962.274,77 €	- €	- €	€ 2.962.274,77	€ 2.962.274,77
Heading 1 -Agencies - EUSPA Govsatcom/SSA		**28.226,00 €**	141,13 €	141,13 €	- €	- €	€ 141,13	€ 141,13
Heading 1 -Agencies - EUSPA H2020/Europe		**22.905.619,62 €**	675.715,78 €	675.715,78 €	137.433,72 €	137.433,72 €	€ 538.282,06	€ 538.282,06
Heading 1 -Agencies - FRONTEX		**7.181.314,24 €**	35.906,57 €	35.906,57 €	- €	- €	€ 35.906,57	€ 35.906,57
Heading 1 -Agencies - EMSA		**11.038.147,92 €**	55.190,74 €	55.190,74 €	- €	- €	€ 55.190,74	€ 55.190,74
Heading 5 -Agencies - EDA		**12.963.193,81 €**	64.815,97 €	64.815,97 €	- €	- €	€ 64.815,97	€ 64.815,97
Heading 1 -Agencies - EEA		**20.991.510,74 €**	104.957,55 €	104.957,55 €	- €	- €	€ 104.957,55	€ 104.957,55
Heading 1 -Agencies - SATCEN -EAS		**4.658.597,22 €**	23.292,99 €	23.292,99 €	- €	- €	€ 23.292,99	€ 23.292,99
Heading 5 -Procurement		**1.121.696,94 €**	5.608,48 €	5.608,48 €	785,19 €	785,19 €	€ 4.823,30	€ 4.823,30
Total Operational & Admin budget		**1.590.874.971,21 €**	9.809.230,67 €	9.809.230,67 €	169.482,19 €	169.482,19 €	9.639.748,49 €	9.639.748,49 €
Operating Budget *(only for Executive Agencies)*		**- €**	- €	- €	- €	- €	€ -	€ -
Total		**1.590.874.971,21 €**	9.809.230,67 €	9.809.230,67 €	169.482,19 €	169.482,19 €	9.639.748,49 €	9.639.748,49 €

DG DEFIS' supervision arrangements are based on the principle of intensive control of the entrusted entity and, for EUSPA, participation in the Administration Board.

The cost of controls is highly outweighed by their benefits. The EU Space Programme components are major industrial programmes of significant size and complexity.

Although the audited samples of financial transactions are not statistically representative, DG DEFIS considers that the error rates detected by its ex-post audits, are a reliable indicator for the non-audited transactions. The level of errors remains relatively stable over the years, so that the error rates detected previously on the 2020 costs for the non-audited international organisations in 2021, can be used as a reliable indicator for the error rate in 2021. In order to mitigate the risk that the actual error rate could be higher on the non-audited entities, DG DEFIS decided to apply a conservative approach and to declare an error rate of 0.50% for the budget delegated to these non-audited organisations in 2021. For more detail on legality and regularity of the transactions, see annex 7.

b. Fraud prevention, detection and correction

DG DEFIS has developed and implemented its own anti-fraud strategy since 2020, on the basis of the methodology provided by OLAF. It was last updated on 01 February 2022 and advertised on DG DEFIS Intranet. Its implementation is being monitored and reported to the management annually. All necessary actions have been implemented. The anti-fraud strategy is an essential element in the development of an anti-fraud culture within the DG. DG DEFIS contributes whenever needed to the actions identified in the Commission Anti-Fraud Strategy and follows up on eventual financial recommendations.

The fraud risk assessment is integrated in the annual risk assessment exercise. As the Directorate externalised the majority of its budget implementation, the anti-fraud strategy has been targeted towards the supervision of the implementation of anti-fraud strategies by the DG DEFIS's entrusted entities.

The controls aimed at preventing and detecting fraud are similar to those intended to ensure the legality and regularity of the transactions. They are effective and no weakness has been identified in 2021. **On the basis of the available information, DG DEFIS has reasonable assurance that the anti-fraud measures in place are overall effective.**

c. Other control objectives: safeguarding of assets and information, reliability of reporting (if applicable)

Reliability of reporting

DG DEFIS delegates most of its budget implementation to external entities. In addition to the pillar assessment of these entities (except EU decentralised agencies), prior to the conclusion of any contribution agreement, it also relies on the declarations of assurance and on the unqualified audit opinions provided every year.

All controls performed by DG DEFIS on ex-ante and ex-post level revealed overall no material misstatements on the accounts presented by the entrusted entities under the programmes managed by DG DEFIS.

Valuation and Safeguarding of assets and information

The total asset value on the balance sheet at end 2021 is EUR 10.729 billion. The non-current assets amount to EUR 8.435 billion of intangible assets, property, plant and equipment and long-term pre-financing. Furthermore, EUR 304 million of current assets consists of pre-financing managed and controlled in the context of the DG's direct and indirect management of space and defence related activities.

The property, plant and equipment balance (EUR 8.128 million) is the most material one in the DEFIS balance sheet given hereafter. It corresponds to the net value of the assets generated by the implementation of the EU space programme (namely: EGNOS, Galileo and Copernicus). By end of 2021, the gross assets values of this programme are:

- EUR 6 413 million for Galileo;
- EUR 4 505 million for Copernicus; and,
- EUR 467 million for EGNOS (European Geostationary Navigation Overlay System).

Further information on these balances are provided in the notes to the balance sheet given in annex 3. The depreciation charge related to these space assets amounted to EUR 625 million for the year 2021.

During 2021, the controls performed on the data provided by ESA and EUSPA for the valuation of the space assets were maintained. Specific workshops on the operational stages of asset development were organised by DG DEFIS to discuss per programme the costs to be capitalised and the stages of operational development. Site inspections, will be resumed once the travel restrictions due to COVID are lifted. In spite of the sanitary situation, EUSPA managed to carry out inspections.

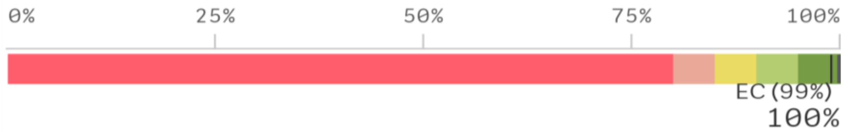
Regarding the Copernicus Satellite – Sentinel 1B, ESA indicated that the instrument was unavailable since 23 December 2021. The anomaly is related to a power supply function required for the radar operations. The technical teams are working on the determination of the root cause and possible work around solutions. ESA informed users of the risk of long-term unavailability of data provision (several months). At this stage, since investigations are still on-going, no impairment has been registered in the accounts for the year 2021. We will see in 2022 if an action should be taken.

2. *Efficiency of controls*

DG DEFIS manages a large portfolio of heterogeneous activities in the defence industry and space sectors, involving different ways of implementation. Based on an assessment of the most relevant key indicators and control results, DG DEFIS has assessed the efficiency of the control system and reached a positive conclusion.

Timely payments:

In 2021, DG DEFIS ensured efficient processing of payments within the legal deadline. The corporate indicator for **timely payments**, reaches 100%.

Timely Payments	DG Score	EC Score
	**100%**	**99%**

Procurements:

The average time to publication in 2021 is relatively high (190 days) due to the complexity for the launch and conclusion of 2 contracts.

Grants:

The high level of the indicators ‘time to inform’ and ‘time to grant’ (mostly related to the EDIDP programme) are due to double comitology, security classification, high complexity of the programme, validation/input from Member States during grant agreement preparation and the need to develop and consolidate new rules and procedures.

DG results for the reporting year		
Key DG indicator on control <u>efficiency</u>	**Grants**	**Procurements**
Complaints received from unsuccessful economic providers	0	0
Number of new cases received by the **Ombudsman** in 2020 relating to grant / procurement procedures	0	0
Number of **legal proceedings** initiated by contractors or economic providers against the Commission relating to grant / procurement procedures	0	0
Number of instances of **overriding of controls** in relation to grant / procurement procedures	0	0
Past due critical and/or very important audit **recommendations**	0	0
Average **time to publication** of selection results (days)		190,0
Coverage of **first level *ex ante*** controls		100 % of all commitments and payments
		100 % of all tender documents and evaluation reports
Coverage of **second level *ex ante*** controls		100 % of all tender documents and evaluation reports
Average time to grant (TTG) (days)	342,0	
Average time to pay (TTP) (days)	18,5	
Average **time to inform** applicants of the outcome of the evaluation of the application (TTI) (days)	213,0	
Average days of **suspension** (days)	9	
Percentage of payments suspended in comparison to all payments executed	12,9%	

All the procedures put in place have been optimised in 2021 to work more efficiently.

3. *Economy of controls*

The cost of controls at Commission level is assessed by the cost of the different control stages. The assessment for each management mode is obtained from the ratio between these costs and the total amount paid in the year for the related management mode. The cost of control for assets is not reported as they are included in the supervision costs of entrusted entities.

Cost of control – direct management

The level of control in direct management was cost-effective in 2021 for both procurement and grant processes.

DG results for the reporting year - *Direct management*		
Common indicators on cost of control	**Grants**	**Procurements**
Percentage of **overall cost of control of grant / procurement process** in comparison to total expenditure executed during the year	3,41%	
	3,30%	3,91%
Overall cost of control (grant/procurement)	€ 3.830.337	€ 1.000.524
Average number of ongoing contracts (grant/procurement) managed per full time equivalent	2,57	7,71
Percentage of costs of control related to the **evaluation and selection** procedure in comparison to the total value of grants / procurement contracted	1,36%	2,3%
Average value of ongoing grant / procurement agreements managed per full time equivalent	€ 15.093.033	€ 4.828.759
Average **project management costs** per ongoing grant / procurement agreement	€ 53.199	€ 12.507
Total Costs of ex post audits	€ 5.700	

Cost of control – indirect management

DG results for the reporting year - *Indirect management*								
Common indicators on cost of control	**International Organisations and other entrusted entities**							
Overall cost of control of supervision process	€ 4.254.216							
Percentage of **overall cost of control** of supervision process in comparison to the total annual amount delegated excluding any remuneration paid	0,17%							
Remuneration fees paid (to international organisations, agencies, EIF)	€ 239.117.030							
	ESA	Eumetsat	ECMWF	Mercator	EEA	Frontex	Satcen	EUSPA
	€ 86.096.730	€ 5.900.000	€ 3.000.000	€ 2.200.000	€ 680.000	€ 405.000	€ 1.498.341	€ 139.336.959
Percentage of cost of **remuneration fees paid** to entrusted entities in comparison to the total annual amount delegated excluding any remuneration paid	9,39%							
Percentage of costs of control related to the **establishment or prolongation** in comparison to the total annual amount delegated	0,02%							
Percentage of costs of control related to the **reporting and subsequent monitoring** of the execution in comparison to all payments executed	0,127%							
Total cost of ex post audits	€ 209.550							

Overall, the cost of monitoring and supervision controls for the implementation of the EU space programme, represents in DG DEFIS 0.17 % of the total annual amount delegated and paid. The cost of controls is highly outweighed by their benefits considering the major responsibilities of DG DEFIS especially in the space sector with the development and management of space assets owned by the EU.

The high percentage of ‘remuneration fees’ paid to international organisations and to entrusted entities is mainly due to the technical specificities of the EU space programme which are major industrial programmes of significant size and diversity. They can be implemented only through large entities such ESA and EUSPA. The control strategy in indirect management is considered to be cost-effective overall.

4. Conclusion on the cost-effectiveness of controls

Based on the most relevant key indicators and control results, DG DEFIS assessed the effectiveness, efficiency and economy of its control system and reached a positive conclusion on the cost-effectiveness of the controls for which it is responsible.

DG DEFIS is of the opinion that the current control system is the best suited for fulfilling the relevant control objectives efficiently and at a reasonable cost. It represents a good balance between the invested efforts (internal control costs and remuneration fees), the obtained error rates (effectiveness of controls) and delivery of objectives (efficiency). The declaration of assurance (Section 2.1.5) includes no reservation for the expenditure categories or control systems concerned.

2.1.2. Audit observations and recommendations

This section sets out the observations, opinions and conclusions reported by auditors, including the limited conclusion of the Internal Auditor on the state of internal control. Summaries of the management measures taken in response to the audit recommendations are also included, together with an assessment of the likely material impact of the findings on the achievement of the internal control objectives, and therefore on management's assurance.

Internal Audit Service (IAS)

The IAS concluded in 2021 their audits on:

- The supervision of the implementation of the 2014-2020 programme for EGNOS (2020): six important recommendations were made for which a satisfactory action plan was proposed and accepted. IAS will follow-up on it during 2022.
- Payments and accounting for tangible assets under the Galileo and Copernicus 2014-2020 programmes (2020) with two important recommendations for which a satisfactory action plan was identified during the audit to mitigate the risks. The execution will be followed up by the IAS during 2022.
- Limited review of the error rate calculation in DG DEFIS (IAS.C2-2021 Y COMM-002). The IAS did not identify any important findings and concluded that the error rates are correctly calculated and that the reporting of the error rates in DG DEFIS’ 2020 AAR is reliable. In its contribution to the 2020 AAR process of DG DEFIS, IAS states that DEFIS management has accepted all the recommendations issued 2019-2021, that it has adopted action plans to address the residual risks identified by the auditors and that it has assessed a number of action plans as implemented which

have not yet been followed up by the IAS. Therefore, IAS concluded that the internal control systems in place for the audited processes are effective.

As foreseen in their action plan of 2021, an audit on the preparedness of DG DEFIS for the management of the European Defence Fund was launched. The fieldwork started in January 2022 and is still ongoing.

European Court of Auditors (ECA)

ECA's Annual Report 2020: On 11 November 2021, the Court adopted its Report on the performance of the EU Budget – Status at the end of 2020. The report mentions in its Chapter 2 - Competitiveness for growth and jobs – the programmes Galileo and EGNOS, without any specific comments for these two programmes.

ECA Statement of Assurance (DAS) 2020: No specific comments related to DG DEFIS.

ECA specific audits in 2021: The ECA launched in September an audit on the Commission Financial Architecture, concerning all DGs but the ECA did not consider DG DEFIS for further contributions. The ECA launched another audit in December about the reliability of the accounts, addressing the 2021 closure. This audit is on-going.

ECA performance audits: The ECA finalised in 2021 its audit started in 2019 on the performance of the uptake of services for Galileo and Copernicus. In July, ECA issued the ECA's Special Report N° 7/2021 "EU space programmes Galileo and Copernicus: services launched, but the uptake needs a further boost", which contains 4 recommendations [1 a)b), 2, 3 a)b)c), 4 a)b)c)]. Only 1a is very important, all the others are marked as important. In addition, in 2021 the ECA started an audit to assess whether PADR had prepared the EU for increasing its spending on defence through the EDF. The preliminary meeting took place in July 2021 and the kick-off of the audit in January 2022.

2.1.3. Assessment of the effectiveness of internal control systems

The Commission has adopted an Internal Control Framework based on international good practice, to ensure the achievement of its policy and management objectives. Compliance with the internal control framework is a compulsory requirement. DG DEFIS uses the organisational structure and the internal control systems suited to achieving its policy in accordance with the internal control principles and due regard to the risks associated with the environment in which it operates.

To achieve its objective, DG DEFIS largely relies on entrusted entities and regulatory agencies, as well as on close cooperation with various partners and international organisations. There are risks associated to the complexity of the EU space programme and to technological uncertainties linked to innovative developments. It is key for DG DEFIS to ensure a tight supervision of entrusted entities for a sound implementation of its programmes. Looking at Defence Industry, the Commission has increased its effort with the adoption of the European Defence Fund, becoming with this programme one of the top three defence research investors in Europe.

Concerning the overall state of the internal control system, DG DEFIS complies with the three assessment criteria for effectiveness, i.e. (a) staff has the required knowledge and skills, (b) systems and procedures are designed and implemented to manage the key risks effectively, and (c) there are no instances of ineffective controls that have exposed the DG to its key risks.

The Head of Unit in charge of Risk Management and Internal Control reported on the state of internal control and provided her opinion to the Director-General. No issues were raised that may affect assurance.

To conclude, DG DEFIS has assessed its internal control system during the reporting year and has concluded that it is effective and the components and principles are present and functioning well overall, but some improvements are needed as minor deficiencies were identified relating to the internal control principles 5 (clearer job descriptions), 9 (improving change management) and 13 (increase filing of documents).

2.1.4. Conclusions on the assurance

The above-mentioned approach provides sufficient guarantees as to the completeness and reliability of the information reported and results in a comprehensive coverage of the budget delegated to the Director-General of DG DEFIS.

In conclusion, based on the elements reported above, management has reasonable assurance that, overall, suitable controls are in place and working as intended; risks are being appropriately monitored and mitigated; and necessary improvements and reinforcements are being implemented. The Director General, in his capacity as Authorising Officer by Delegation has signed the Declaration of Assurance Declaration of Assurance.

2.1.5. Declaration of Assurance

I, the undersigned,

Director-General of DG DEFIS,

In my capacity as authorising officer by delegation

Declare that the information contained in this report gives a true and fair view ([63]).

State that I have reasonable assurance that the resources assigned to the activities described in this report have been used for their intended purpose and in accordance with the principles of sound financial management, and that the control procedures put in place give the necessary guarantees concerning the legality and regularity of the underlying transactions.

This reasonable assurance is based on my own judgement and on the information at my disposal, such as the results of the self-assessment, ex-post controls, the work of the Internal Audit Service and the lessons learnt from the reports of the Court of Auditors for years prior to the year of this declaration.

Confirm that I am not aware of anything not reported here which could harm the interests of the institution

Brussels, 31 March 2022

e-signed

Timo Pesonen

([63]) True and fair in this context means a reliable, complete and correct view on the state of affairs in the DG/Executive Agency.

2.2. Modern and efficient administration – other aspects

2.2.1. Human resource management

A first minor reorganisation took place mid-March 2021, creating a HR Business Correspondent cell reporting to the Director General, and a second reorganisation in May 2021, creating a new unit in the Directorate A to better respond to the implementation needs of the European Defence Fund. The HRBC team implemented all HR processes, monitored the measures linked to the COVID-19 crisis, and ensured the necessary staffing reinforcement, notably for the Defence Industry Directorate, via seconded national experts from national defence administrations.

The local HR strategy is being developed within DG DEFIS with the goal to present it in 2022 after staff consultation.

In the pandemic context, DG DEFIS organised in 2021 regular on-line staff meetings for the whole DG to present the key challenges and objectives of the DG and maintain an efficient staff awareness of key priorities on space and defence matters.

DG DEFIS **internal communication** strategy focused on maintaining close links with staff in order to keep them connected and adequately informed. This was done via corporate digital tools and with the DG DEFIS Buzz weekly newsletter, weekly video recordings of "DG DEFIS' Management Meeting Debriefing", a 'Question Box" series presenting staff in a more informal way, and monthly '"DEFIS Lab" webinars with external experts to stimulate common exchanges, brainstorming and out-of-the-box thinking on topical issues of relevance to both Defence Industry and Space.

2.2.2. Digital transformation and information management

Under a Memorandum of Understanding (MoU) signed between DG DEFIS and DG GROW[64], the DG DEFIS Information Resource Manager (IRM) worked closely with DG GROW to implement the Commission data governance and principles.

DG GROW provided support to DG DEFIS in its implementation of the document management policy of the Commission. This included for instance the improvement of document management practices inside DG DEFIS, as well the creation of its own filing plan, ensuring quality control and the organisation of trainings for keeping staff up-to-date.

In the context of Defence and Space, confidentially and security of information are of utmost importance and must be preserved. To that end and in close cooperation with other DGs, DG DEFIS focused on the re-use of existing tools such as the **eGrant Suite**[65] and to secure operations with the following corporate IT tools which are under development:

([64]) Applicable as from 1 January 2020.

([65]) As part of EC Digital Strategy Action Plan on Reusable Solutions Platform

1) **Exchange of EU confidential information (SUE),** a corporate initiative that will develop a classified IT system.
2) **E-CERTIS system** (owned by DG GROW): an online mapping service for criteria, issuers and evidence in the EU, to be used for future defence procurements.
3) **CERTIDER** (owned by DG TRADE): register for certified defence-related enterprises.

In 2021, the shared **Data Protection** Coordinator (DPC) for DEFIS/GROW replied to a call by the Data Protection Officer in the third trimester to proceed to a review and update of all existing records of processing operations in the Data Protection Management System (DPMS), as well as an update of all related privacy statements. DG DEFIS contributed to the implementation of the Commission's Data Protection Action Plan through the following actions:

- awareness raising activities to promote corporate initiatives, i.e. the international Data Protection Day on the 28 January 2021;
- organisation of unit 'meet your DPC' sessions introducing this function in the DG;
- a dedicated training for senior management (GROW/DEFIS) with the Commissions' Data Protection Officer who presented the general legal framework, the relevance of the rules to the day-to-day business and the different roles and corresponding responsibilities;
- advisory support for units was provided to improve compliance with obligations, i.e. for the creation or update of records and privacy statements, as well as advice on the conformity of IT systems, IT projects and websites with data protection rules and in particular when transferring personal data to third countries or using international cloud services;
- DG DEFIS continuously assess its processing activities in light of the requirements of the Schrems II judgment (C-311/18) and ensures that its processing operations comply with EU data protection law.

2.2.3. Sound environmental management

The DG EMAS correspondents (ECOR) actively participated in network meetings. In close partnership with OIB and DG GROW with which DG DEFIS shares its building, EMAS corporate campaigns and measures were promoted in DG DEFIS to help reducing the environmental footprint of DG DEFIS.

Following the results of a DG DEFIS study "Analysis of the environmental impact of the EU Space Programme" to assess its footprint and environmental impact specific actions were designed regarding further developments of life cycle environmental footprint of European space activities and sector-specific environmental footprint methodology. Taking into account the complexity of the space value chain coupled with the particular structure of the space industry leads to the need of starting discussions among key stakeholders, gathering industry, institutions and research entities to establish a shared understanding of the most advanced LCA practices, potentially available materials in the space sector and way forward. During 2021, DG DEFIS prepared the framework to start discussions with stakeholders beginning in 2022 with the ultimate aim of implementing a sector-specific category rules (e.g. PEFCR) for the EU space sector activities.

2.2.4. Security of information

All personnel in DG DEFIS are either in possession of a valid security clearance or they have started the vetting procedure. A tailor-made awareness and training plan fitting the needs of DG DEFIS was established by September 2020 by the Security Task Force and continued to be implemented in 2021.